

Are Cryptos More Extreme? The Extreme Behavior of Cryptocurrency Prices

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Abstract

This thesis examines how extreme price movements in cryptocurrencies compare to traditional financial assets like stocks, gold, bonds, and FX. Real market data and straightforward statistical tools are used to study the main features of cryptocurrency returns, such as high volatility, fat tails, volatility clustering, skewness, and market correlations. The presented analysis shows that cryptocurrencies, especially Bitcoin and Ethereum, are much more volatile and prone to large swings than traditional assets. Therefore, the methods employed for analysing traditional assets should be adjusted accordingly when applied to cryptocurrencies, and methods that specialise in modelling extreme phenomena should also be considered (e.g. EVT and GARCH). We also examine how simple models frequently underestimate the risk of extreme price fluctuations in cryptocurrencies, whereas methods from Extreme Value Theory are more effective in capturing these rare events. Thus, we also demonstrate that implementing Extreme Value Theory in the risk management process for these assets can potentially yield more accurate estimates. Overall, the findings presented aim to highlight the importance of choosing the right models for understanding and managing risk in cryptocurrency markets.

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1 Introduction

Over the last decade, cryptocurrencies have become increasingly popular, driven by growing institutional interest, technological advancements and a surge in speculative trading fueled by media coverage and global economic uncertainty. With their growing popularity, it is important to understand the fundamental principles behind this unique asset class.

Cryptocurrencies are digital assets secured by cryptography, and they are typically maintained through decentralised, distributed ledgers that are called blockchains. The process by which most cryptocurrencies are created and validated is known as mining. Mining involves participants, known as miners, using computational power to solve complex, mathematical puzzles that secure the network and confirm transactions (Nakamoto, 2008). This process is central to the consensus mechanisms of most cryptocurrencies, with Proof-of-Work (PoW) being the pioneering protocol. In PoW systems, miners compete to solve cryptographic challenges, and the first to solve the challenge appends a new block to the blockchain and receives a reward in the form of newly minted cryptocurrency and transaction fees (Nakamoto, 2008). This mechanism not only issues new coins but also deters fraudulent activity by making it computationally expensive to alter the transaction history.

The first cryptocurrency, Bitcoin, was introduced in 2008 by Satoshi Nakamoto in the white paper, “Bitcoin: A Peer-to-Peer Electronic Cash System” (Nakamoto, 2008). Bitcoin’s core technological innovation is its use of a distributed ledger (the blockchain) and consensus through PoW, and also a fixed supply cap of coins. The technology behind Bitcoin inspired a new class of programmable blockchains, the most notable being Ethereum. Launched in 2015, Ethereum extended the blockchain framework by enabling decentralised applications and smart contracts, which are self-executing agreements coded directly on the blockchain (Buterin, 2014). While Bitcoin is focused primarily on digital money, Ethereum functions as both a cryptocurrency (Ether, ETH) and a global, decentralised computing platform (Wood, 2025).

The mining of cryptocurrencies originally happened on ordinary personal computers. However, as Bitcoin and Ethereum networks grew, mining became dominated by specialised hardware, known as ASICs (Application-Specific Integrated Circuits) for Bitcoin, and high-end GPUs for Ethereum (prior to Ethereum’s transition to Proof-of-Stake in 2022) (Wood, 2025). The majority of mining activity has historically been concentrated in regions with abundant and inexpensive electricity, most notably China (before its 2021 ban on cryptocurrency mining), followed by the United States, Russia, and Kazakhstan. These locations offer economic incentives for miners due to low energy costs and, in some cases, more lenient regulations.

Expert opinion on cryptocurrency mining is divided. Proponents argue that mining underpins blockchain security, fosters innovation, and enables financial inclusion. Critics highlight the environmental impact of PoW mining, particularly the vast energy consumption and carbon emissions associated with large-scale mining farms (Stoll et al., 2019). The debate has led to a growing interest in alternative consensus mechanisms (such as Proof-of-Stake), which Ethereum adopted in its 2022 “Merge” upgrade to reduce energy consumption significantly (CCRI, 2022).

2 Do cryptocurrencies exhibit statistically significant higher extreme price movements than traditional assets?

2.1 Stylized Facts of Cryptocurrencies

To provide context, the stylized facts of the major asset classes—equities, fixed income, currencies, cryptocurrencies, and commodities—are first summarised. Particular emphasis is placed on the stylized facts of cryptocurrencies, with comparisons drawn to the other asset classes wherever similar characteristics are present.

1. **Extreme volatility:** Cryptocurrencies, especially Bitcoin and alternative coins (altcoins), display significantly higher volatility than traditional assets. The annualised volatility of Bitcoin is around 4 times higher than that of equities and gold (4.03 for the S&P500 and 4.08 for Gold), and it is worth noting that altcoins are even more volatile than Bitcoin or Ethereum.

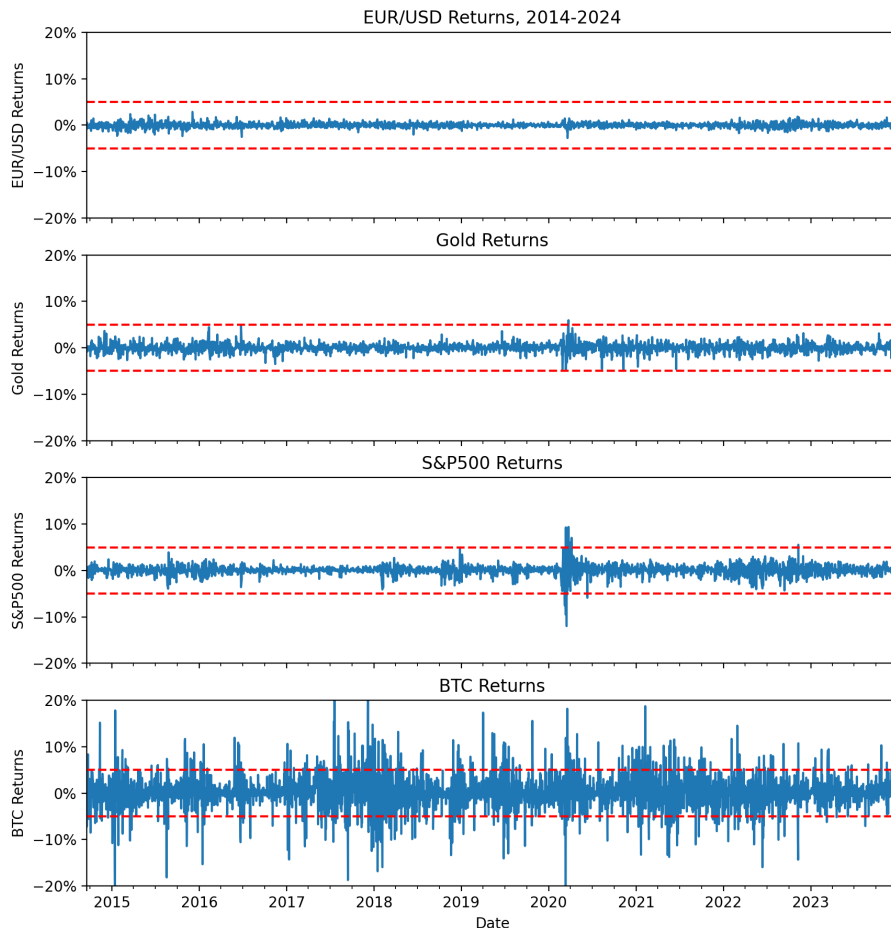


Figure 1: Returns Compared

Asset	Daily	Monthly	Quarterly	Yearly
S&P500	1.10%	5.06%	8.76%	17.53%
BTC	3.70%	20.26%	35.08%	70.65%
Gold	1.09%	5.00%	8.67%	17.34%
EUR/USD	0.72%	3.30%	5.72%	11.44%
Bonds	0.33%	1.50%	2.61%	5.21%

Table 1: Annualized Volatility (%) Across Different Assets and Frequencies

2. **Fat-Tailed Distributions:** Similarly to equities and commodities, cryptocurrencies have heavy-tailed return distributions, which is proven by their high kurtosis values. It is important to highlight that the fat tails of cryptocurrency returns are often stronger, which translates to more frequent extreme price movements. Kurtosis is a statistical metric that measures how heavy or light are the tails of an observed distribution compared to a normal distribution, and it is a crucial metric in risk management. Using Fisher’s definition, a kurtosis value of greater than zero means more extreme outliers, and less than zero signals fewer outliers which means less extreme movement (value of zero means that it is like a normal distribution). The analysis yields excess kurtosis values of 11.60 for Bitcoin and 10.85 for Ethereum (using return data between 2015-2024) which indicate that these assets exhibit extremely fat-tailed behaviour.

Excess Kurtosis (BTC):	14,8400
Excess Kurtosis (ETH):	12,2284

Table 2: Excess Kurtosis for BTC and ETH

3. **Volatility Clustering:** Similarly to both equities and currencies, cryptocurrencies experience periods of high volatility that are often followed by even higher volatility. Both Bitcoin’s and Ethereum’s volatility is persistent, as previously shown, they have statistically significant fat tails, therefore GARCH (Generalized Autoregressive Conditional Heteroskedasticity) models provide a good fit for modeling the observed volatility.

To begin, the presence of volatility clustering in the time series data can be assessed by performing an ARCH test which checks for autoregressive conditional heteroskedasticity (ARCH). The test yields p-values lower than 0.05 for both Bitcoin and Ethereum, which is significant evidence for heteroskedasticity.

ARCH Effect (BTC): p-value =	1,316E-18
ARCH Effect (ETH): p-value =	2,544E-27

Table 3: ARCH Effect Test p-values for BTC and ETH

Next, the analysis of heteroskedasticity is extended by fitting a GARCH model to the rescaled log-returns. For both examined cryptocurrencies, the p-values of

alpha and beta are lower than 0.05, and moreover, the sum of the alpha and beta values are very close to one, which implies the persistence of conditional volatility, and indicates significant volatility clustering. This means that, for example, a shock to Bitcoin returns can cause volatility to remain elevated for a long time.

BTC GARCH(1,1) p-values:			ETH GARCH(1,1) p-values:		
ω :	0.0176199	<0.05	ω :	0.0695797	>0.05
α :	0.0313799	<0.05	α :	0.0043069	<0.05
β :	0.0000000	<0.05	β :	0.0000000	<0.05
$\alpha + \beta$ (BTC): 0.9638			$\alpha + \beta$ (ETH): 0.9869		
→ close to 1: strong persistence			→ close to 1: strong persistence		

Table 4: GARCH(1,1) p-values and persistence for BTC and ETH

4. **Asymmetric Returns:** To continue, the asymmetry of the distribution of returns can be inspected by calculating the skewness. Using Fisher’s definition of skewness for our selected cryptocurrencies, the values are -0.79 for Bitcoin and -0.93 for Ethereum respectively, which implies asymmetric risk for these cryptos. Negative skewness implies a higher probability of extreme negative returns occurring than positive ones, therefore the risk of a large sudden drop is always present. Also, negative skewness signals non-normal behaviour and therefore implementing traditional VaR models for cryptos can lead to inaccurate estimates. It is also worth noting that Bitcoin is less negatively skewed than Ethereum, which can be due to Ethereum’s smaller market capitalization and its higher sensitivity to regulatory news.

Skewness (BTC):	-0,7936
Skewness (ETH):	-0,9318

Table 5: Skewness of BTC and ETH Returns

5. **Highly Non-Stationary Returns:** Cryptocurrency returns have the interesting tendency of trending for long periods, this property is called non-stationarity. Another important fact to highlight is that, unlike commodities for example, cryptocurrencies don’t have a fundamental “anchor”. Therefore, because of these special properties, momentum trading strategies may be potentially implemented. This property of crypto can be checked by performing a mean reversion test, more specifically an Augmented Dickey-Fuller (ADF) test. The null hypothesis of the ADF test is that a unit root in the time series exists, which means stationarity, and the alternative hypothesis is that the unit root doesn’t exist, which in turn means non-stationarity.

To put it into context, a stationary time series has constant mean and variance over time while non-stationary time series often display trends and increasing variance. When performing the ADF test, a p-value lower than 0.05 means that

the series is stationary, and a value larger than 0.05 means that the series is non-stationary. After performing the ADF test on both Bitcoin and Ethereum, the obtained p-values are 0.62 for Bitcoin and 0.58 for Ethereum, both of them are larger than 0.05 which implies non-stationary returns for both cryptocurrencies, and therefore no mean-reversion (tendency of the series to return to its average level over time).

ADF Test (BTC): p-value =	0,6156	(Stationary if < 0,05)
ADF Test (ETH): p-value =	0,5789	(Stationary if < 0,05)

Table 6: ADF Test Results for BTC and ETH

6. **High Correlations Within Crypto Markets:** To continue, another stylized fact that should be taken into account is the correlation of cryptocurrencies. As an example, Bitcoin and Ethereum are highly correlated, with a correlation value of 0.79, which is in sharp contrast with the correlation between Bitcoin and the S&P500 Stock Index, which is only -0.02. Altcoins are also highly correlated with each other, the Ethereum-BinanceCoin pair having a historical correlation of 0.67 and Ethereum-Solana pair having 0.63. Another interesting property of Bitcoin, is that its correlation with equities fluctuates over time.

Bitcoin-S&P 500 Correlation:	-0,0234
Bitcoin-Ethereum Correlation:	0,7947
Bitcoin-BinanceCoin Correlation:	0,6675
Bitcoin-Solana Correlation:	0,5502
Ethereum-BinanceCoin Correlation:	0,6650
Ethereum-Solana Correlation:	0,6286
BinanceCoin-Solana Correlation:	0,5604

Table 7: Pairwise Correlations Among Crypto Assets and S&P 500

7. **Weekend and Overnight Effects:** A significant difference between cryptocurrencies and equities is that cryptocurrencies trade 24/7. Contrary to claims in earlier literature and financial commentary regarding significant weekend and overnight effects in cryptocurrency markets, the obtained results, using the Friedman test on Bitcoin, Ethereum, Solana, and Binance Coin, do not support the presence of a statistically significant weekend effect in the analysed period. The Friedman test is a nonparametric method typically used to detect differences in distributions across multiple related groups, which makes it suitable for comparing return distributions across weekdays. While this method provides an initial check for systematic differences, it may not fully account for time-series dependencies such as autocorrelation. Therefore, more robust approaches (e.g., regression models with weekday dummy variables) and analysis at higher frequencies or within specific subperiods could potentially reveal patterns that are more consistent with earlier findings.

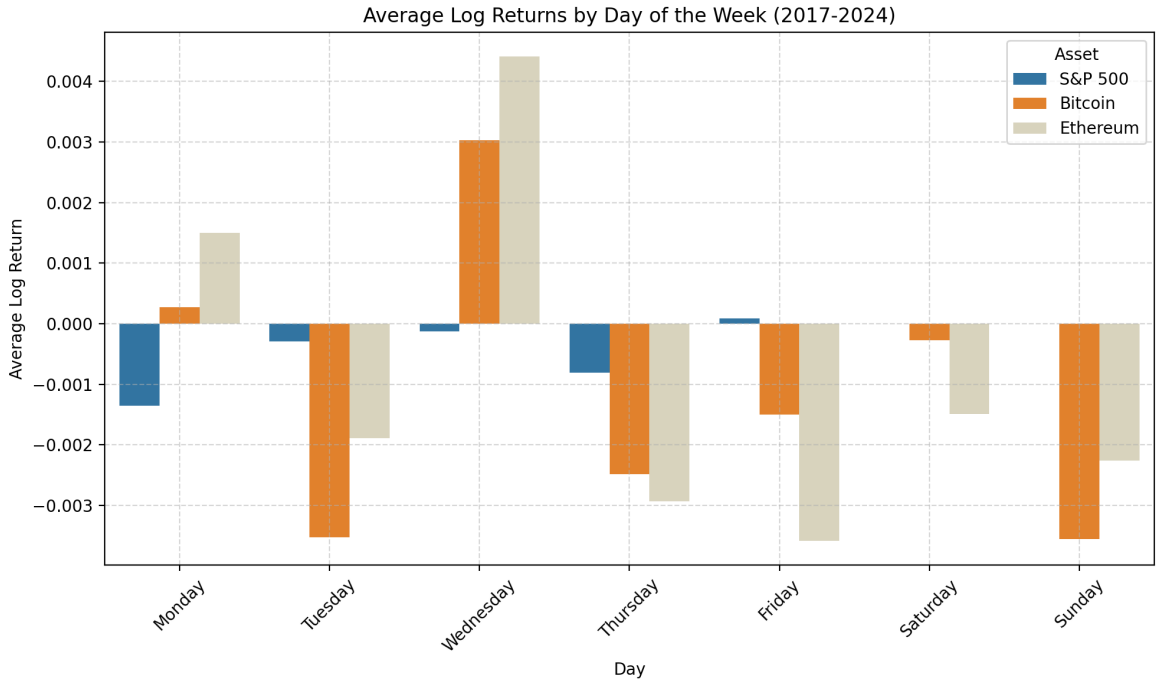


Figure 2: Average Log Returns Compared

Day of the Week	Bitcoin (%)	Ethereum (%)	S&P500 (%)
Monday	0.03%	0.15%	-0.14%
Tuesday	-0.35%	-0.19%	-0.35%
Wednesday	0.30%	0.44%	0.30%
Thursday	-0.25%	-0.29%	-0.25%
Friday	-0.15%	-0.36%	0.01%
Saturday	-0.03%	-0.15%	—
Sunday	-0.36%	-0.23%	—

Table 8: Average Daily Returns (%) by Day of the Week for Bitcoin, Ethereum, and S&P500

Asset	Friedman Statistic	p-value
BTC	8.9029	0.1791
BNB	6.0092	0.4222
SOL	5.4511	0.4874
ETH	9.3962	0.1525

Table 9: Friedman Test Results

8. **More Stylized Facts:** Last but not least, another important factor to highlight with cryptocurrencies is that unlike equities, cryptos are fragmented across multiple exchanges. This is important to consider because this can lead to price deviations between exchanges, which in turn can create arbitrage opportunities. For Altcoins that have a lower capitalization, slippage and illiquidity can also be present.

Also, cryptos are unique in the sense that they have no intrinsic valuation models. Stocks have earnings, bonds have yields, commodities have supply and demand, but cryptocurrencies have no widely accepted fundamental valuation model, they are driven by speculation, which can lead to strong momentum trends and bubbles.

It is also worth mentioning that Bitcoin (and other cryptos as well) has halving events which can trigger bull markets, but for example, Ethereum is not exposed to supply shocks like that because it uses staking instead of mining.

Comparison with Other Assets				
Stylized Fact	Equities	FX	Commodities	Crypto
Fat Tails	Yes	Yes	Yes	Extreme
Volatility Clustering	Yes	Yes	Yes	Stronger
Mean Reversion	No	No	Yes (oil)	No
Skewness	Negative	Slightly Negative	Mixed	Varies
Correlation with Equities	Medium	Low	Varies	Changing
Weekend Effect	No	No	No	Diminished

Table 10: Stylized Facts: Comparison Across Asset Classes

2.2 Connection between Extreme Value Theory and cryptocurrencies

After reviewing the stylized facts about cryptocurrencies and traditional financial assets, this section explores how Extreme Value Theory (EVT) serves as a useful tool for analyzing the behaviour of cryptocurrencies, particularly in the context of extreme market events.

As established above, cryptocurrency returns often exhibit heavy-tailed distributions and EVT is designed to model such tail risks. Since cryptos are more volatile than traditional assets like stocks, bonds and commodities, the application of EVT could help quantify the likelihood and severity of extreme events exhibited by cryptos. EVT can also prove useful from a risk management point of view since standard models like VaR using Gaussian assumptions underestimate the tail risk of cryptocurrencies thus EVT could provide a more accurate risk estimate in the tails of the distribution. It is also a more convenient statistical method since it doesn't assume an underlying distribution for the whole dataset, rather, it only models the tails where the extreme events happen.

Within this statistical approach, there are two main methods for identifying and extracting the extreme values: the Block Maxima Method and the Peaks Over Threshold Method. Their underlying procedure for extreme selection is quite different, but when implementing extreme value analysis, the two approaches should be used together in order to get a more detailed result.

2.2.1 Block Maxima Method

To begin, this method essentially divides the time series data into blocks and then selects either the maximum or minimum value from each block. After the extraction of the extreme values the Generalized Extreme Value (GEV) distribution is fitted to the selected extremes. It is very important to highlight that for this method, the number of extracted extreme values are directly proportional to the block size that we select, and therefore the model fit will also directly depend on how we choose the block size. For the POT method, this aspect is different because for the extraction of the extremes both a threshold and a de-clustering value have to be defined, therefore it is more difficult to predict how many datapoints will be obtained in the end as a result.

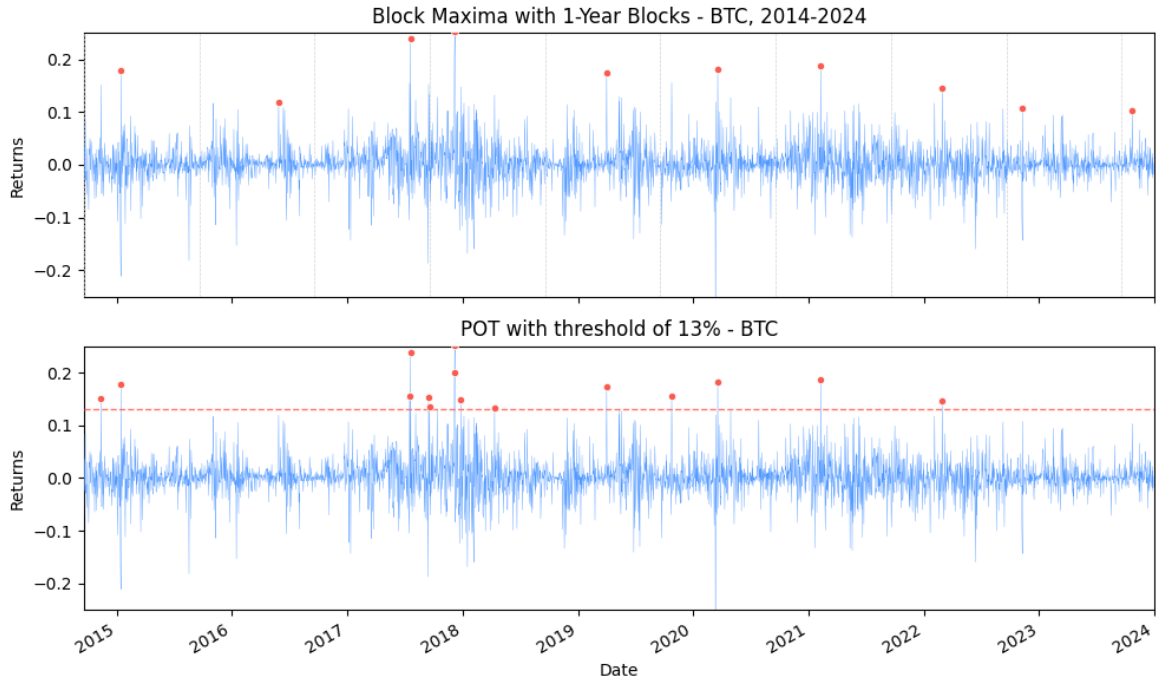


Figure 3: BM and POT Method for BTC, 2014-2024

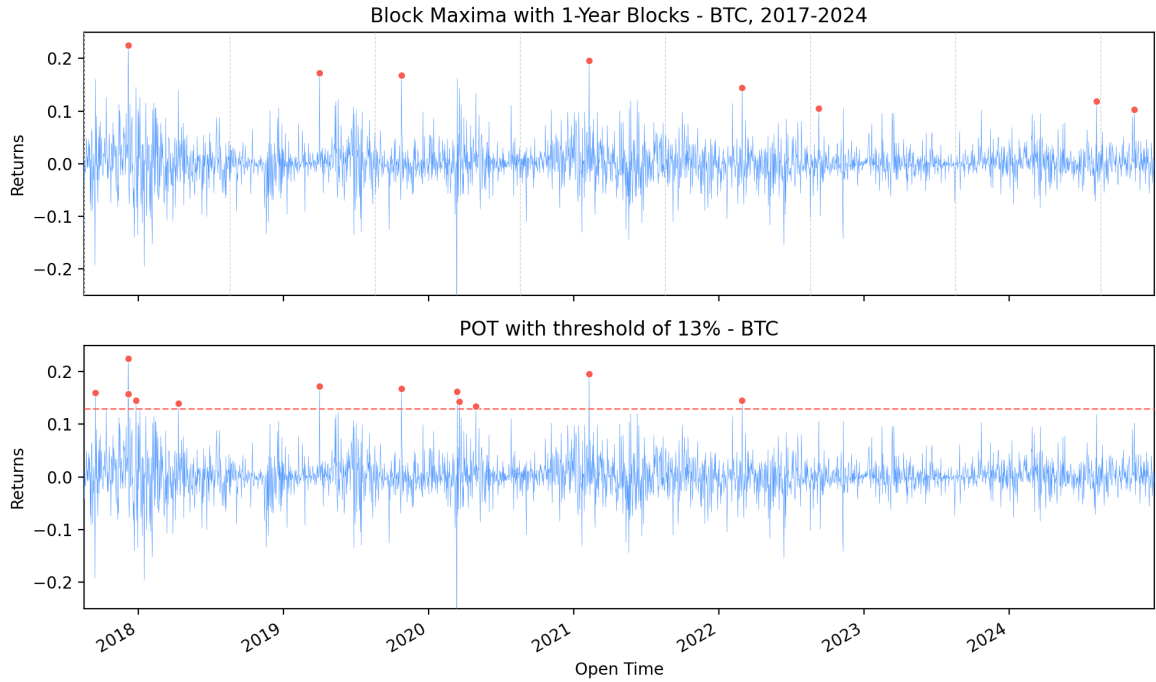


Figure 4: BM and POT Method for BTC, 2017-2024

The difference between the two methods is illustrated using historical Bitcoin returns over two distinct time horizons. Since yearly blocks are set for the BM Method, exactly 10 values will be extracted during 2014-2024 and 8 values during 2017-2024. On the other hand, with a set threshold of 13%, the POT method extracts 15 values for 2014-2024 and 12 values for 2017-2024. Moreover, POT seems to de-cluster the extreme returns less effectively than the BM method around the year 2018.

As the next step, extreme Bitcoin returns for the period 2017–2024 were extracted using the Block Maxima (BM) method. After testing multiple block sizes, a four-day block was selected based on the stability and consistency of the results. Following the extraction, the GEV distribution was fitted to the obtained values and diagnostic plot with 95% confidence level was generated.

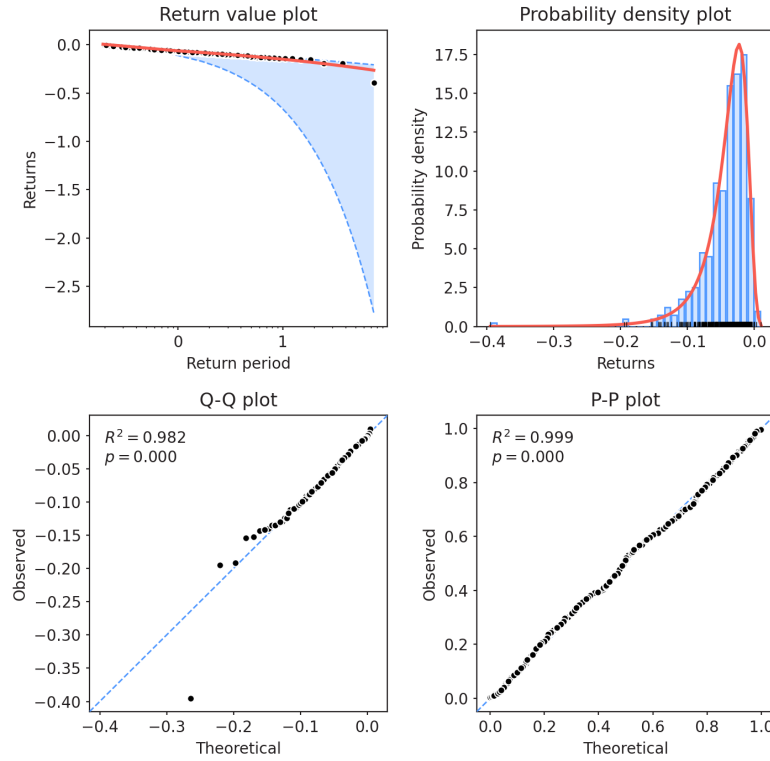


Figure 5: BTC Block Maxima Diagnostic Plot, 2017-2024

The GEV distribution fits the extracted values quite accurately. The P-P Plot which describes the overall distributional fit indicates that the GEV distribution fits the overall dataset quite well. While the Q-Q plot displays a good fit as well, it tries to capture issues in the tail. The diagnostic yields $R^2 = 0.982$, which suggests a high degree of correlation between the theoretical and observed quantiles. The small deviations compared to the 45-degree line in the lower tail suggest that there are a couple of extremes that are more severe than what the theoretical distribution predicted. The red line in the Return Value plot depicts the fitted GEV model and the blue region resembles the 95% confidence interval. Since the black dots (the actual return values from our sample) mostly follow the red GEV line, the model fit is relatively good. The blue region (confidence interval) captures most of the points therefore the model's confidence intervals are realistic. If any dots would fall below the shaded region, that would indicate that some extreme losses are more severe than the model predicted.

In addition, various probability distributions were fitted to the extracted extremes and compared visually to evaluate their fit. Among them, the GEV distribution demonstrated the highest alignment with the observed values.

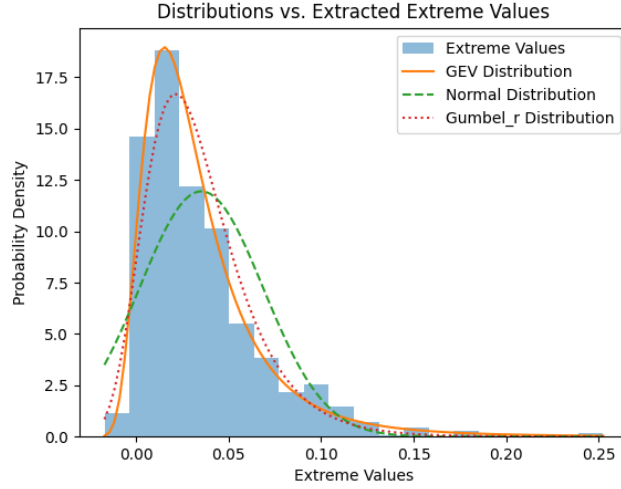


Figure 6: Bitcoin Overlaid Distributions

2.2.2 Block Minima Method

Similar results are obtained when the block minima method is applied to the dataset of Bitcoin returns from 2017 to 2024. The following graphic displays the extracted extreme values using 4 day blocks.

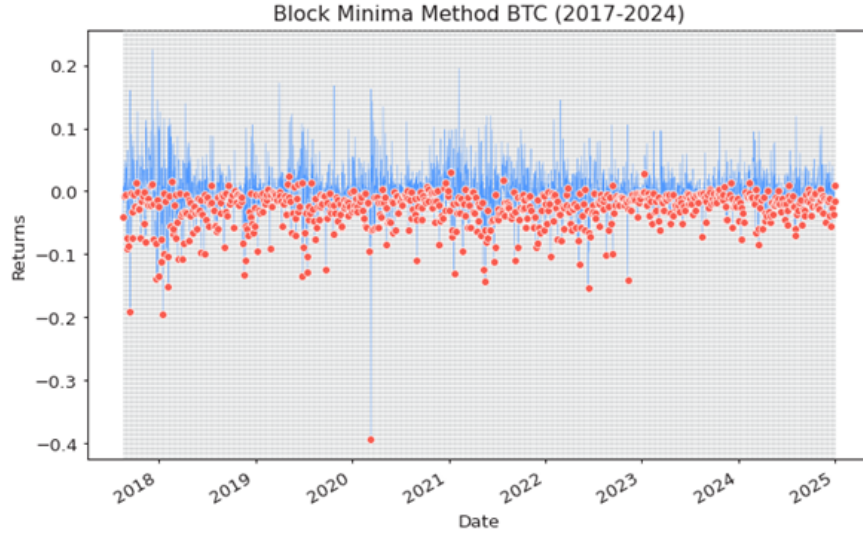


Figure 7: Bitcoin Block Minima Method

After fitting the GEV distribution to the extracted extremes that are now the extreme losses, the model fit is checked again by running a diagnostic with a 95% confidence level. When looking at the probability density plot in the diagnostic, the GEV distribution (red line) underestimates the occurrence of extreme losses since some observed extreme losses are more severe than what the GEV distribution predicted. This is also seen in the Q-Q plot, where the R^2 is very high which indicates a good overall model fit, but there are some eye-catching deviations compared to the 45-degree line, in the lower tail there are a couple of extremes that are way more severe than what the

theoretical distribution predicted. Nevertheless, the return value plot confirms that the confidence interval captures all the extreme losses but some deviate from the theoretical distribution (red line) quite considerably. And last but not least, the P-P plot displays a very high R^2 and indicates a very good model fit.

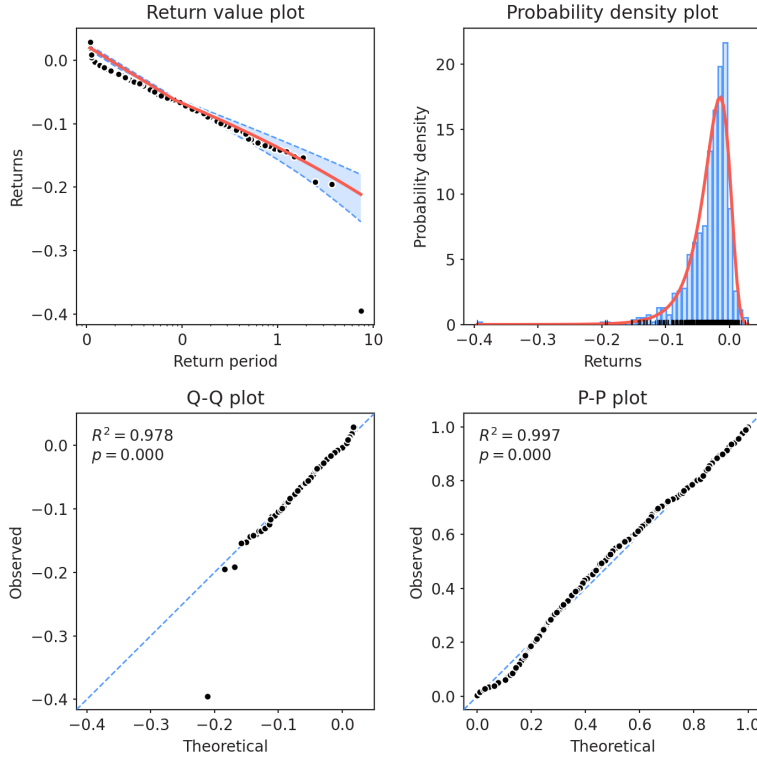


Figure 8: BTC Block Minima Diagnostics

The different probability distributions can be overlayed again on the extracted extreme values to check which one fits the observations the most appropriately. Again, the GEV distribution fits the data the best out of the three overlayed distributions, but again it is worth noting that the GEV distribution underestimates the extremity of the historical losses of Bitcoin.

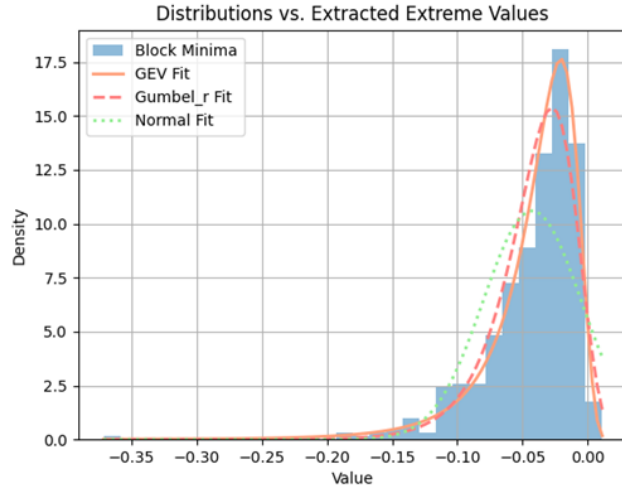


Figure 9: BTC Overlaid Distributions, Block Minima

2.2.3 Peaks Over Threshold Method

As the next step in the EVT analysis of Bitcoin, the Peaks over Threshold method is used to extract the extreme values and then the Generalized Pareto Distribution is fitted to the obtained extremes. For the POT method 7 days was chosen as the de-clustering value and 10% as the threshold value. In the graphic below, it is clearly seen that significantly less datapoints are obtained (only 29) compared to when using the BM method so it can be expected that the overall model fit probably will not be that good.

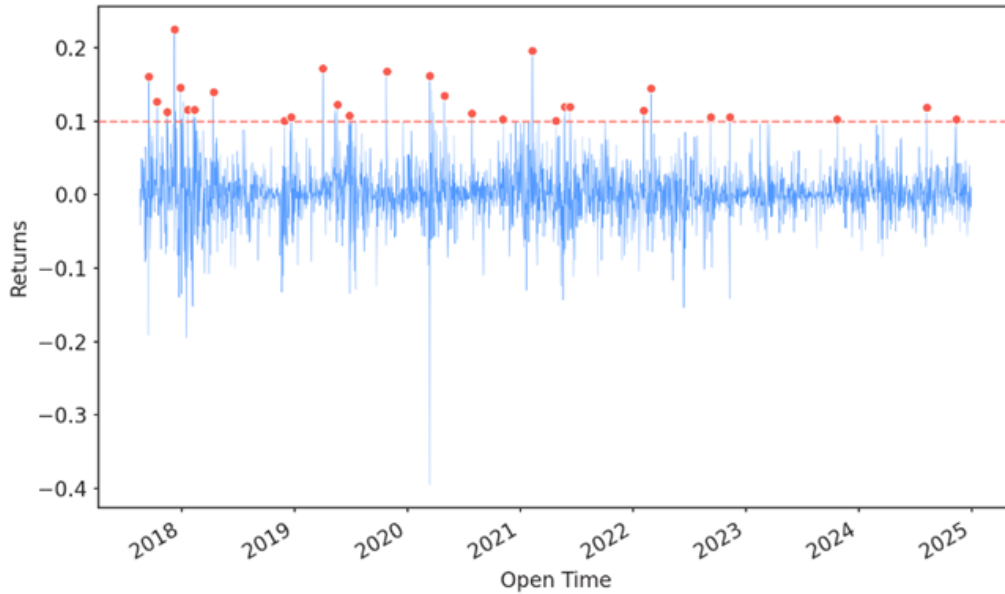


Figure 10: Bitcoin Peaks Over Threshold

After the extraction, the Generalized Pareto Distribution is fitted to the obtained data points and a diagnostic of the model with a 95% confidence interval is generated. It is clear that the model fit is not that excellent, but the same pattern arises as with the Block Maxima method. On the probability density plot, the datapoints follow the theoretical distribution (GPD, red line) relatively accurately. The return value plot confirms this tendency and none of the datapoints lie outside the confidence interval (blue region) which indicates good model fit, but the points don't lie that closely to the 45-degree line either. The P-P plot indicates a good fit with a high R^2 with datapoints slightly deviating from the line of perfect agreement between the fitted distribution and the observed data. The Q-Q plot shows that in the upper tail of the distribution of the observed data there are a couple of extremes that are more severe than what the theoretical distribution predicted, but still the datapoints follow the 45-degree line quite accurately.

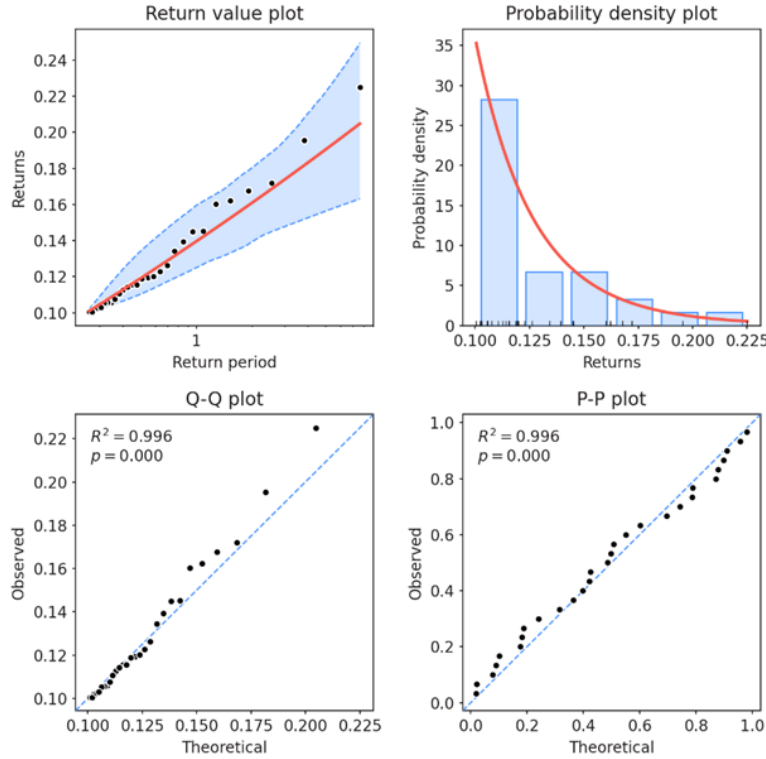


Figure 11: Bitcoin Peaks Over Threshold Diagnostic

The GPD model fit is further analysed by overlaying it with other distributions. It is important to add that by default, the GPD assumes $\text{loc}=0$, but since the chosen threshold was 10%, this should be reflected in the distribution, so the GPD curve (red line) was shifted by 0.1 to the right by adjusting the loc parameter. Both the GPD and exponential distribution fit the observed data quite well but not too smoothly.

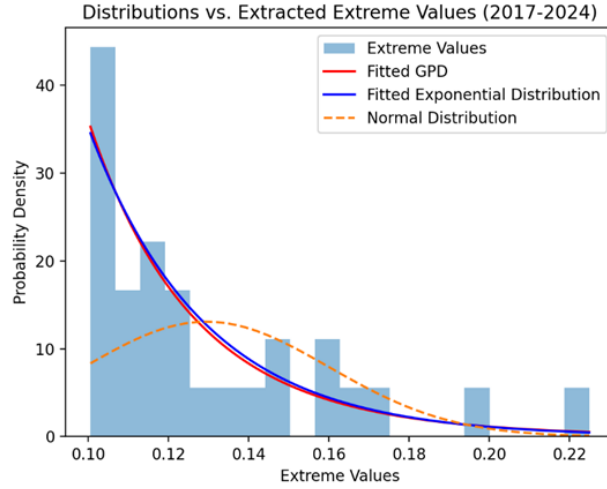


Figure 12: Bitcoin Overlaid Distributions

2.2.4 Peaks Below Thershold

As a continuation, the analysis of the Bitcoin losses was carried out with the Peaks Below Threshold method as well, with 7 days set as the de-clustering value and 10% as the threshold value for the 2017-2024 time period. Not too many datapoints are obtained here (only 23) with these settings either, but very significant outliers can be noticed.

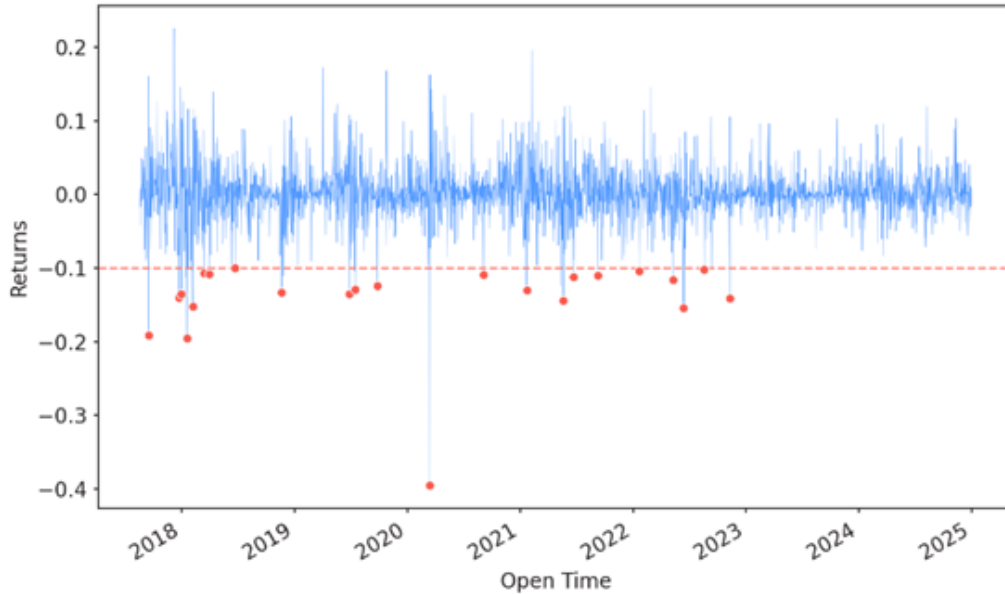


Figure 13: Bitcoin Peaks Below Threshold

The diagnostic summary was generated again of the model fit at the 95% confidence

level and due to the scarcity of observations the R^2 values are significantly lower than in the previous diagnostics. In the Q-Q plot a huge outlier can be observed which is from the year 2020, but other than that, the rest of the points follow the agreement line quite accurately. Also, in the return value plot no datapoint lies outside the confidence interval and on the probability density plot the observations roughly follow the shape of the GPD.

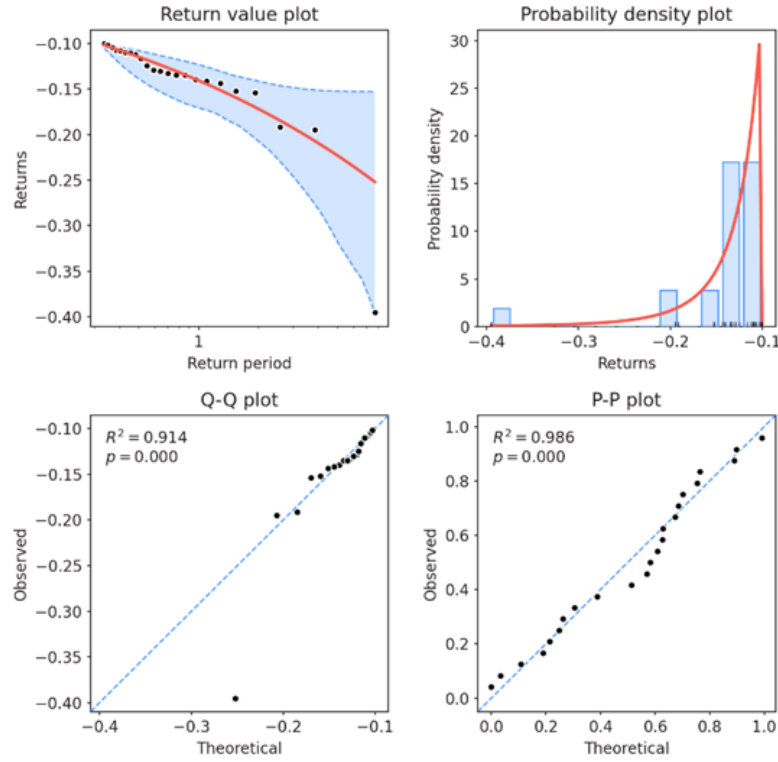


Figure 14: BTC Peaks Below Threshold Diagnostics

Even though the data coverage is not that good, after overlaying the distributions, the observations tend to follow the GPD line (orange).

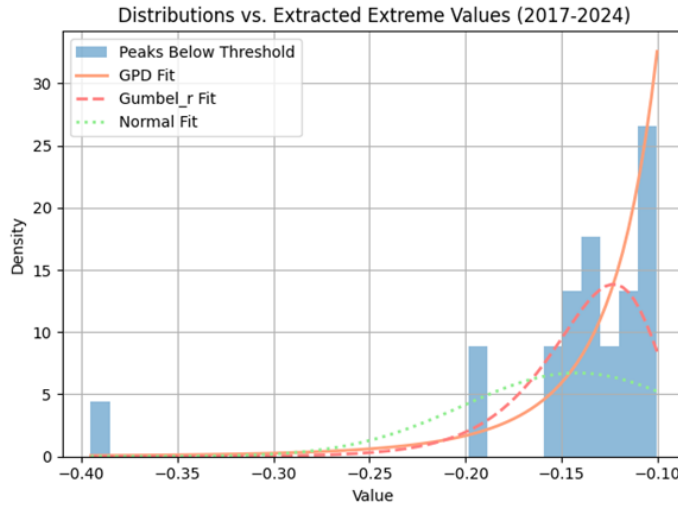


Figure 15: BTC Peaks Below Threshold Overlaid Distributions

2.2.5 Goodness of Fit Results

In order to quantify the how well each probability distribution fits the extracted extreme values, three goodness of fit tests were performed, namely: Kolmogorov-Smirnov Test, Cramér-von Mises Test, Anderson-Darling Test. All three tests compare the empirical distribution with a theoretical distribution function, but they give more weight to different characteristics. Their main features are summarised in the following table.

Test	Focus	Most Sensitive To	Tail Sensitivity	Use Case Example
Kolmogorov-Smirnov	Max deviation	Location & scale shifts	Low	General-purpose distribution check
Cramér-von Mises	Overall deviation	Whole-distribution mismatch	Moderate	Detecting subtle general deviations
Anderson-Darling	Weighted deviation	Tails and global mismatch	High	Tail-heavy risk modeling (e.g., EVT)

Table 11: Comparison of Distribution Fit Tests

All the test results were collected in a table for both EVT methods. Since there were more datapoints to work with when executing the Block Maxima approach, the results are clearer. According to the K-S test and the Cramér-von Mises test the observed data follows a GEV distribution which the A-D test doesn't confirm but rejects the normal and right-skewed Gumbel distribution as possibilities.

For the POT method, the results are not that definitive but when looking at the D-value for each test which aims to quantify the difference between the theoretical and the empirical distributions, a clear pattern can be noticed. Even though neither the K-S nor the Cramér-von Mises test reject the exponential distribution, its D-value is significantly larger than that of the Generalized Pareto Distribution.

In conclusion, the analysis of the obtained return data of Bitcoin confirms that when applying the block maxima approach, the limiting distribution of block maxima is described by the GEV distribution, encompassing the Gumbel, Fréchet, and Weibull families depending on the shape parameter. Moreover, for the Peaks Over Threshold method in EVT, the corresponding theoretical distribution used to model the exceedances above the threshold is the Generalized Pareto Distribution.

Test Name	KS Statistic / D Value	p- value	Alpha	Critical Value	Test Result
Kolmogorov-Smirnov Test	0.029600	0.4383	0.05	0.0464	data follows genextreme distribution
Kolmogorov-Smirnov Test	0.068000	0.0007	0.05	0.0464	data doesn't follow gumbel_r distribution
Kolmogorov-Smirnov Test	0.113700	5.3096	0.05	0.0464	data doesn't follow norm distribution
Cramér-von Mises Test	0.204306	0.2593	0.05	NA	data follows genextreme distribution
Cramér-von Mises Test	4.965080	0.0000	0.05	NA	data does not follow norm distribution
Cramér-von Mises Test	1.015494	0.0023	0.05	NA	data does not follow gumbel_r distribution
Anderson-Darling Test	29.546026	NA	0.05	0.7830	data does not follow norm distribution
Anderson-Darling Test	80.288911	NA	0.05	0.7520	data doesn't follow gumbel_r distribution

Table 12: Goodness-of-fit results, Block Maxima

Test Name	KS Statistic / D Value	p- value	Alpha	Critical Value	Test Result
Kolmogorov-Smirnov Test	0.061400	0.9937	0.05	0.2028	data follows genpareto distribution
Kolmogorov-Smirnov Test	0.120700	0.5186	0.05	0.2028	data follows expon distribution
Kolmogorov-Smirnov Test	0.220800	0.0254	0.05	0.2028	data doesn't follow norm distribution
Cramér-von Mises Test	0.024791	0.9908	0.05	NA	data follows genpareto distribution
Cramér-von Mises Test	0.136023	0.4365	0.05	NA	data follows expon distribution
Cramér-von Mises Test	0.609461	0.0207	0.05	NA	data does not follow norm distribution
Anderson-Darling Test	12.473143	NA	0.05	1.3230	data does not follow expon distribution
Anderson-Darling Test	3.405769	NA	0.05	0.7290	data does not follow norm distribution

Table 13: Goodness-of-fit results, Peaks Over Threshold

2.2.6 Gold

As a continuation, it is interesting to examine the same properties of gold from the perspective of Extreme Value Theory as it was done with Bitcoin. It is intriguing to compare the two assets since gold produces no direct cash flows, like Bitcoin, but is traditionally perceived as a store of value because of its historical role as money, durability, and universal acceptability across cultures and time periods. Like Bitcoin, gold is not subject to inflationary manipulation, and its physical properties—resistance to corrosion, divisibility, and scarcity—make it suitable as a medium of wealth preservation.

Moreover, the comparison of the two assets is currently at the centre of discussion, since Bitcoin is being considered as a reserve alternative. For example, President Donald Trump signed an executive order establishing a Strategic Bitcoin Reserve, utilizing billions of Bitcoin seized from criminal and civil cases. This reserve is managed by the Treasury and is intended as a long-term asset.

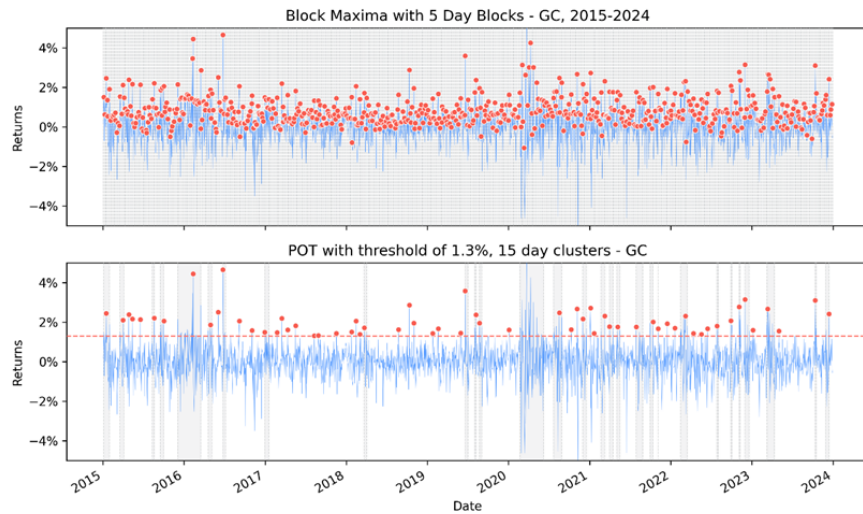


Figure 16: Gold BM and POT

As the first step, the BM method is implemented on the gold returns. Since with there is better data coverage, extreme values were extracted between 1975 and 2024 and thus 601 datapoints were obtained in total.

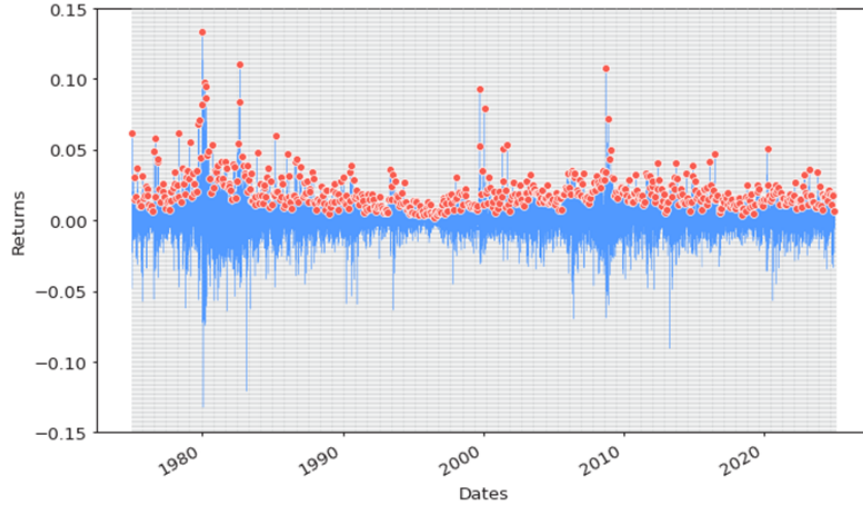


Figure 17: Gold Block Maxima Method

After the extraction, the diagnostic with 95% confidence interval was executed again on the model fit. The model fit is very good based on the return value and probability density plots. The P-P plot confirms that the centre part of the empirical distribution fits very well to the theoretical distribution, which is the GEV distribution in this case, since we are using the BM method. However, the Q-Q plot indicates that the theoretical distribution doesn't predict that well the extreme events in the upper tail of the distribution of our observations. Other probability distributions can be overlayed on the observations and it is observed that the GEV distribution fits well, but so does the Right-Skewed Gumbel Distribution.

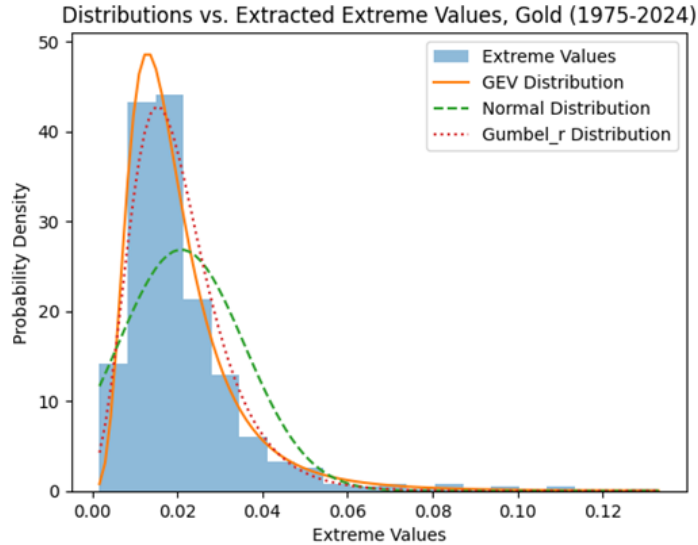


Figure 19: Gold Overlaid Distributions

Next, the behaviour of extreme gold returns can be checked with the POT method as well. With the POT method 128 extremes were extracted, with de-clustering size set to 30.42 days and threshold at 2%.

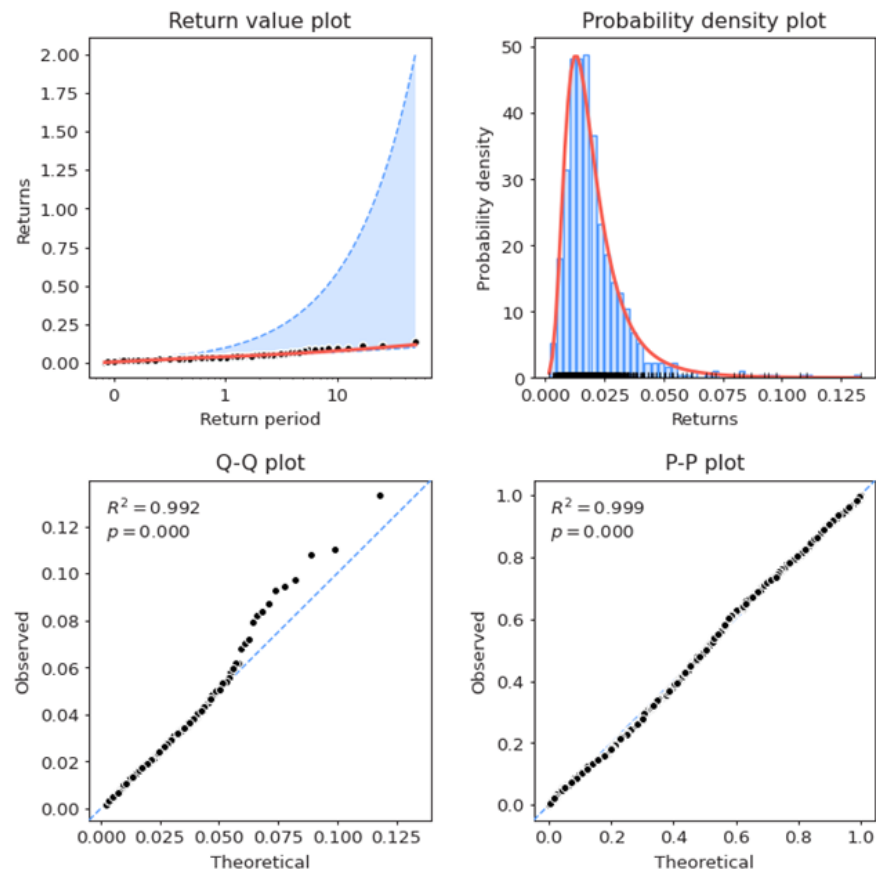


Figure 18: Gold Block Maxima Method Diagnostics

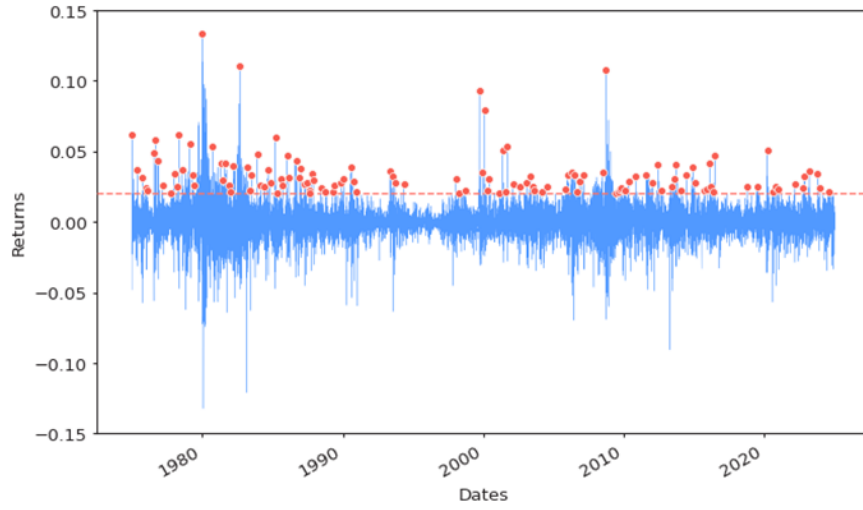


Figure 20: Gold Peaks Over Threshold

The diagnostic with 95% confidence level shows similar results to the one from the BM method, but it seems like the model cannot predict the extreme events that accurately.

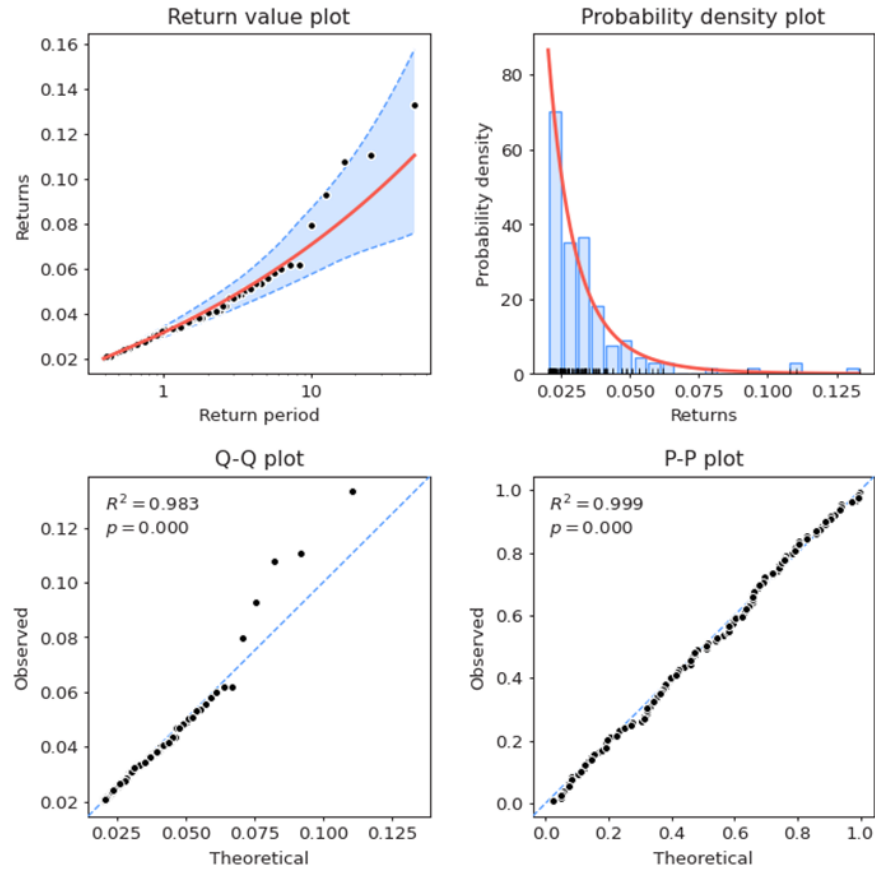


Figure 21: Gold Diagnostics POT

Last but not least, the results of the goodness-of-fit tests can be analyzed, and they are indeed very similar to the results obtained for Bitcoin.

Test Name	KS Statistic / D Value	p- value	Alpha	Critical Value	Test Result
Kolmogorov-Smirnov Test	0.028900	0.6869	0.05	0.0551	data follows genextreme distribution
Kolmogorov-Smirnov Test	0.072600	0.0034	0.05	0.0551	data doesn't follow gumbel_r distribution
Kolmogorov-Smirnov Test	0.159700	7.3391	0.05	0.0551	data doesn't follow norm distribution
Cramér-von Mises Test	0.103139	0.5699	0.05	NA	data follows genextreme distribution
Cramér-von Mises Test	5.795320	0.0000	0.05	NA	data does not follow norm distribution
Cramér-von Mises Test	0.687181	0.0137	0.05	NA	data does not follow gumbel_r distribution
Anderson-Darling Test	33.354334	NA	0.05	0.7820	data does not follow norm distribution
Anderson-Darling Test	89.062222	NA	0.05	0.7510	data doesn't follow gumbel_r distribution

Table 14: Goodness-of-fit results, Block Maxima

Test Name	KS Statistic / D Value	p- value	Alpha	Critical Value	Test Result
Kolmogorov-Smirnov Test	0.047700	0.9191	0.05	0.1187	data follows genpareto distribution
Kolmogorov-Smirnov Test	0.065800	0.6131	0.05	0.1187	data follows expon distribution
Kolmogorov-Smirnov Test	0.219000	7.1486	0.05	0.1187	data doesn't follow norm distribution
Cramér-von Mises Test	0.036127	0.9526	0.05	NA	data follows genpareto distribution
Cramér-von Mises Test	0.173129	0.3265	0.05	NA	data follows expon distribution
Cramér-von Mises Test	1.977922	0.0000	0.05	NA	data does not follow norm distribution
Anderson-Darling Test	25.666025	NA	0.05	1.3350	data does not follow expon distribution
Anderson-Darling Test	11.284897	NA	0.05	0.7640	data does not follow norm distribution

Table 15: Goodness-of-fit results, Peaks Over Threshold

3 How do tail risk characteristics differ between cryptocurrencies and traditional financial markets?

3.1 Behaviour of the left tail

Understanding the behaviour of the left tail of return distributions is essential when assessing risk in financial markets, particularly in the context of extreme losses. The left tail captures rare but severe negative returns, which are critical for risk management, portfolio construction, and regulatory considerations. In the case of cryptocurrencies, whose volatility and market structure differ significantly from traditional financial assets, analyzing tail risk offers valuable insight into their resilience and potential for sudden crashes.

To conduct the analysis of the left tail of each asset multiple steps are followed. First, return values below the 5% and 1% quantiles are extracted. After isolating these specific return values, both the GPD and the Normal distributions are fitted to these log return values, for the GPD the distribution is fitted to the specific exceedances that are below the threshold. For the Normal Distribution different steps are followed, since this distribution type is not designed specifically to capture the tails, but rather the whole distribution. Thus, the Normal Distribution is fitted to all the returns and then its simulated tail behaviour is compared to the one simulated by the GPD. After fitting both distributions to the returns, the fitted parameters are stored and used to simulate the tail behaviour so that it can be compared which distribution estimates the tail behaviour better. Lastly, one sample t-tests are performed to check if the two distributions provide a reasonable fit to the empirical exceedance mean.

3.1.1 Bitcoin

Table 16: GPD Tail Comparison Summary for BTC (5% tail)

Statistic	Value
Threshold (q_5)	-0.076016
Empirical Mean (simple return)	-0.122180
GPD Mean (simple return)	-0.133492
Normal Mean (simple return)	-0.097831
Shape parameter (ξ)	0.7324
Scale parameter (σ)	0.0317
t-statistic GPD	-1.6027
p-value GPD	0.1102
t-statistic Normal	5.1381
p-value Normal	0.0000
GPD exceedances used	194
Normal exceedances used	194

T-test fit to <i>empirical mean</i>?	
GPD	Yes (GPD mean fits empirical mean)
Normal	No (Normal mean differs from empirical mean)

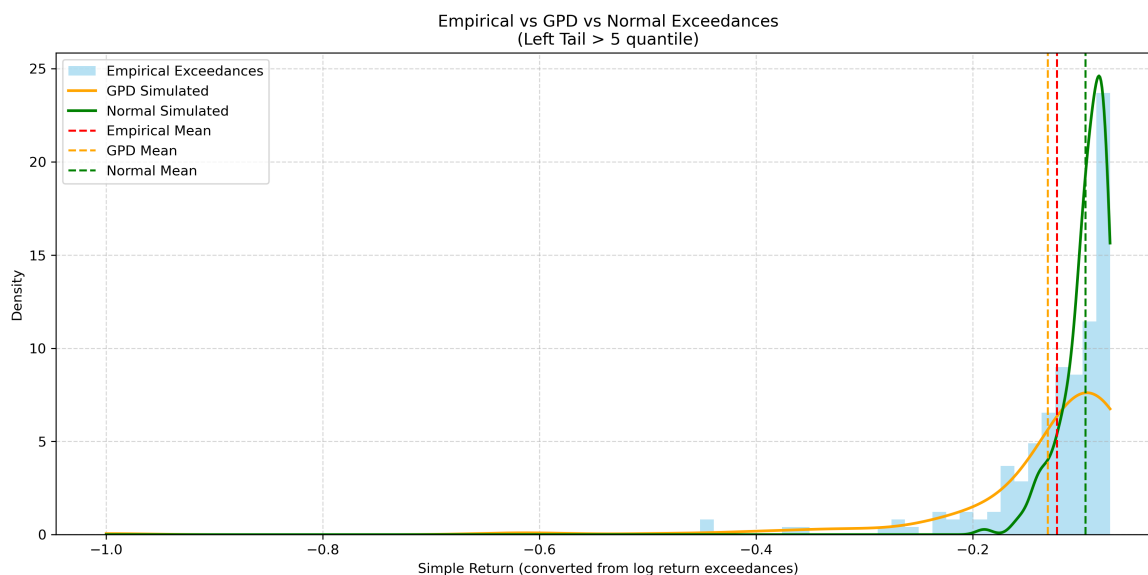


Figure 22: Bitcoin Tail Behaviour, 5% Percentile

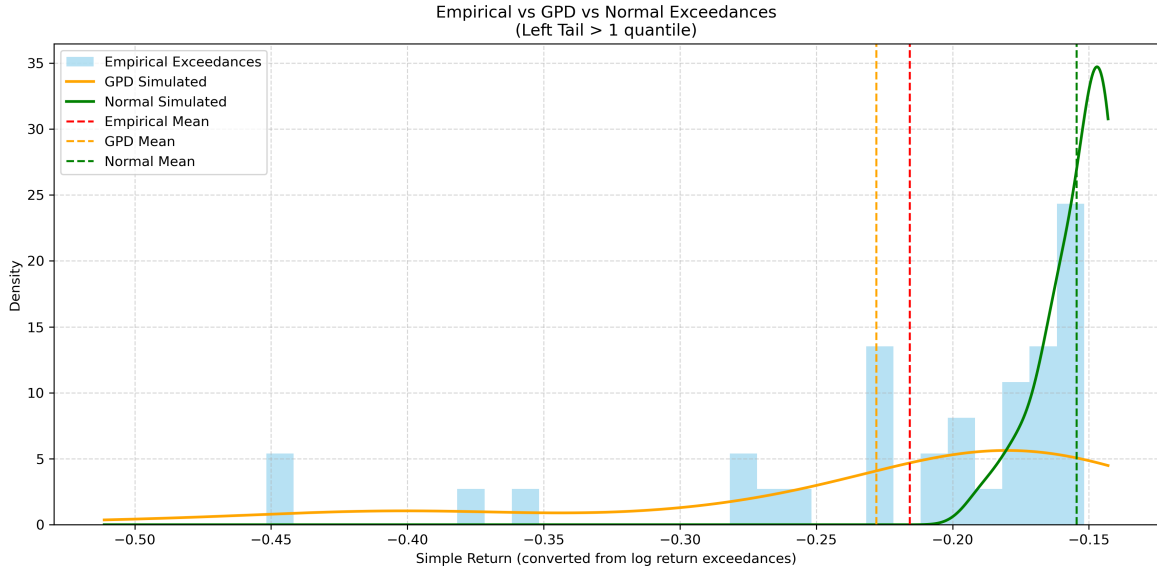


Figure 23: Bitcoin Tail Behaviour, 1% Percentile

3.1.2 Ethereum

Table 17: GPD Tail Comparison Summary for ETH (5% tail)

Statistic	Value
Threshold (q_5)	-0.080812
Empirical Mean (simple return)	-0.116675
GPD Mean (simple return)	-0.115658
Normal Mean (simple return)	-0.094545
Shape parameter (ξ)	0.1543
Scale parameter (σ)	0.0371
t-statistic GPD	0.1684
p-value GPD	0.8664
t-statistic Normal	4.6058
p-value Normal	0.0000
GPD exceedances used	95
Normal exceedances used	95
T-test fit to <i>empirical mean</i>?	
GPD	Yes (GPD mean fits empirical mean)
Normal	No (Normal mean differs from empirical mean)

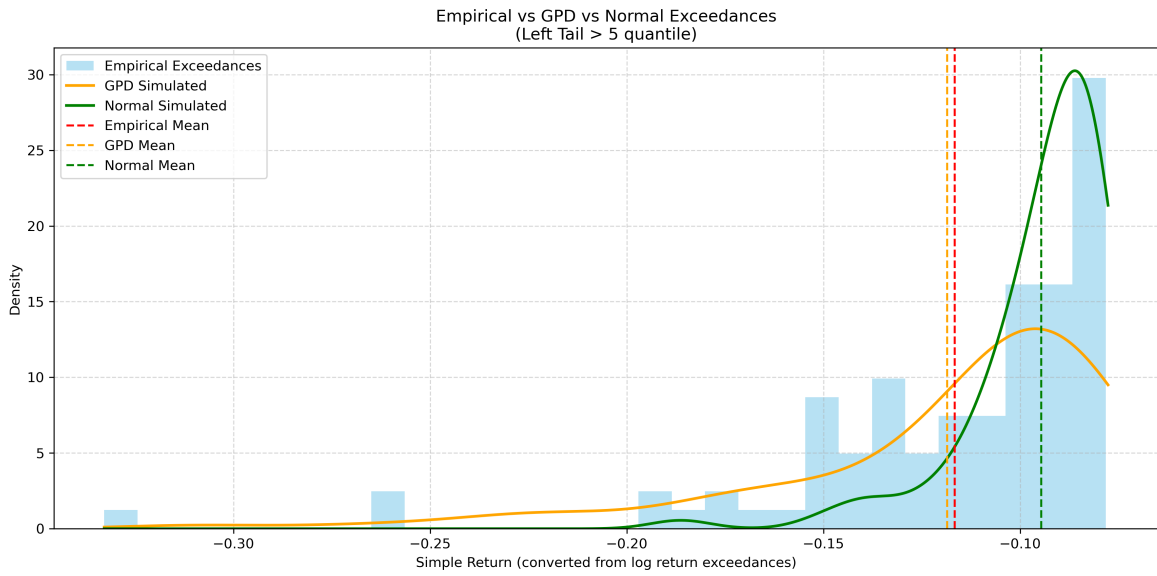


Figure 24: Ethereum Tail Behaviour, 5% Percentile

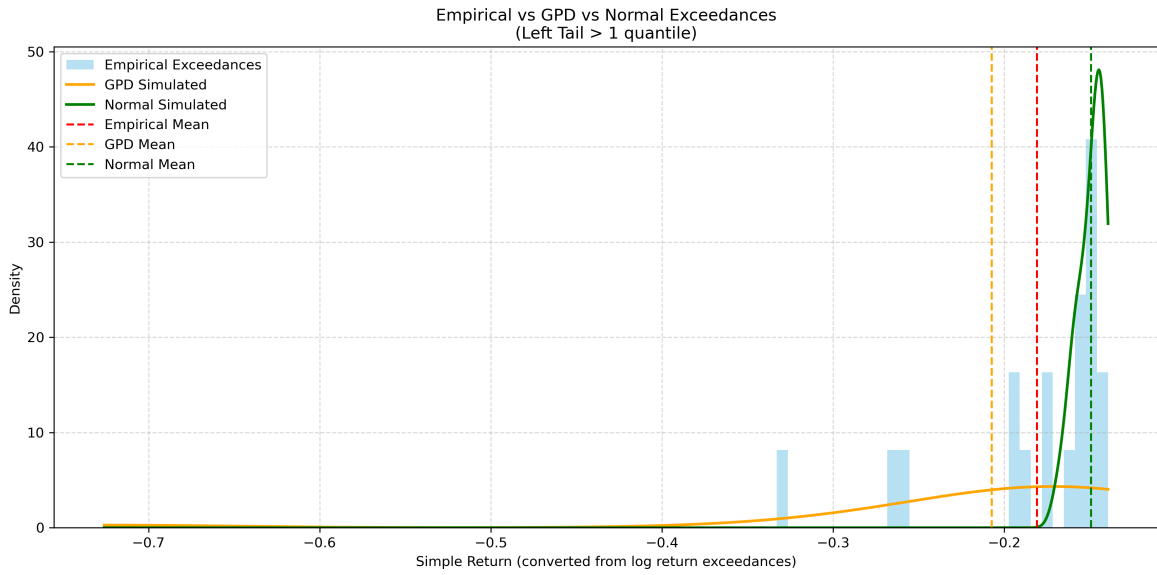


Figure 25: Ethereum Tail Behaviour, 1% Percentile

3.1.3 Gold

Table 18: GPD Tail Comparison Summary for Gold (5% tail)

Statistic	Value
Threshold (q_5)	-0.015422
Empirical Mean (simple return)	-0.022528
GPD Mean (simple return)	-0.024572
Normal Mean (simple return)	-0.018875
Shape parameter (ξ)	0.5631
Scale parameter (σ)	0.0047
t-statistic GPD	-1.5923
p-value GPD	0.1124
t-statistic Normal	5.4607
p-value Normal	0.0000
GPD exceedances used	200
Normal exceedances used	200

T-test fit to <i>empirical mean</i>?	
GPD	Yes (GPD mean fits empirical mean)
Normal	No (Normal mean differs from empirical mean)

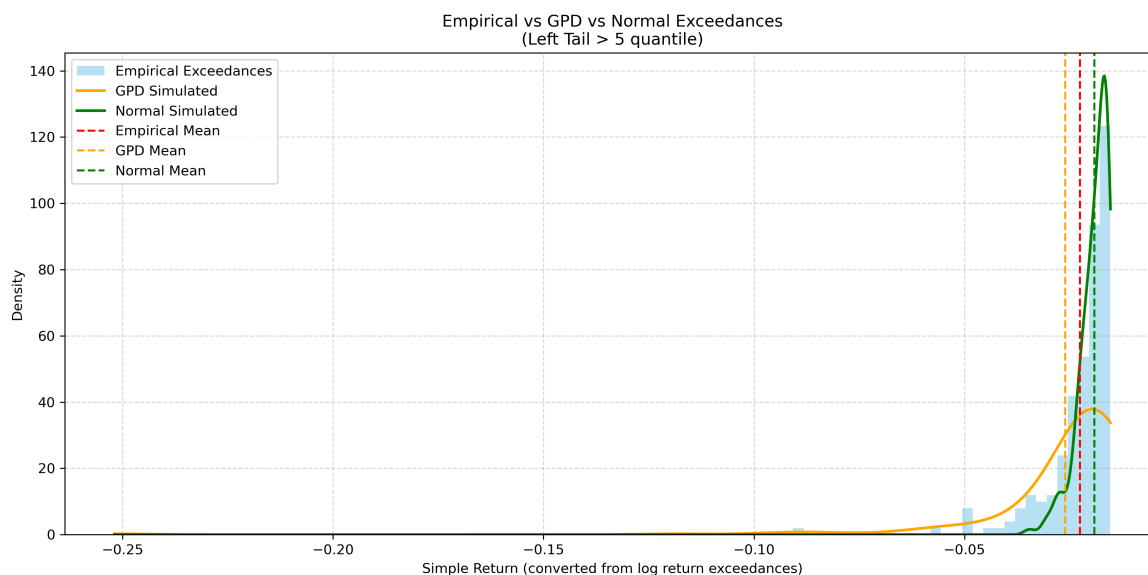


Figure 26: Gold Tail Behaviour, 5% Percentile

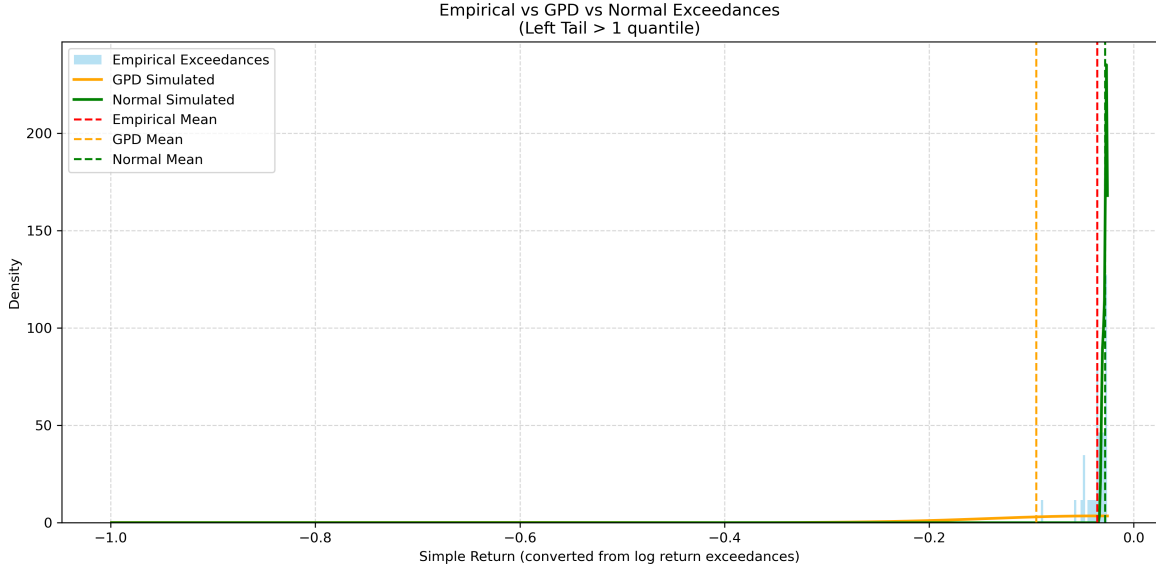


Figure 27: Gold Tail Behaviour, 1% Percentile

3.1.4 Comparing the results

Bitcoin and Ethereum: When looking at the results obtained for Bitcoin, it can be noticed that for the 5% quantile, the GPD provided a much better mean estimate than the normal distribution which significantly underestimated the empirical mean. The t-test also reflects this, since the obtained result says that statistically speaking, the GPD mean fit the empirical mean and the Normal Distribution mean differed from the empirical mean. Furthermore, when looking at the 1% percentile, the GPD mean remains quite accurate, and the gap widens between the normal and the empirical mean, as it would be expected.

In the case of Ethereum, a very similar pattern is present, but the accuracy of the GPD mean estimate was even higher. The GPD mean estimated the empirical mean with a very high precision, and the normal mean severely underestimated it. When looking at the 1% percentile, the GPD mean overestimated the empirical mean, but was still closer to it than the normal mean.

Gold: Compared to Bitcoin and Ethereum, gold paints a unique picture in terms of the applicability of EVT. The t-test of the fitting for the 5% percentile values concludes that the GPD fits while the normal differs from the empirical mean, and the mean estimates also confirm this. Moreover, for gold, the graph for the 1% percentile is quite different, since the GPD mean significantly overestimates the empirical mean, while the normal mean estimate is quite close to it.

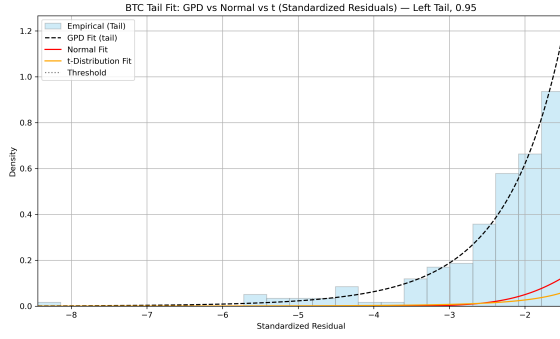
3.2 Combining GARCH with EVT

As it was demonstrated in the stylized facts section, financial time series often exhibit asymmetry, fat tails and volatility clustering. In order to model this volatility clustering behaviour, GARCH(q,p) models can be utilised, where GARCH stands for Generalized AutoRegressive Conditional Heteroscedasticity. The q and p parameters of the GARCH model, refer to history of the return and the volatility of the time series (Jafari et al., 2007). The q parameter refers to the number of autoregressive lags and the p parameter relates to the amount of moving average lags (Jafari et al., 2007). In contrast with the ARCH(q) process, the GARCH(q,p) process allows for lagged conditional variances to enter, and thus incorporates a kind of adaptive learning mechanism, which makes it a very useful tool in the analysis of financial time series (Bollerslev, 1986).

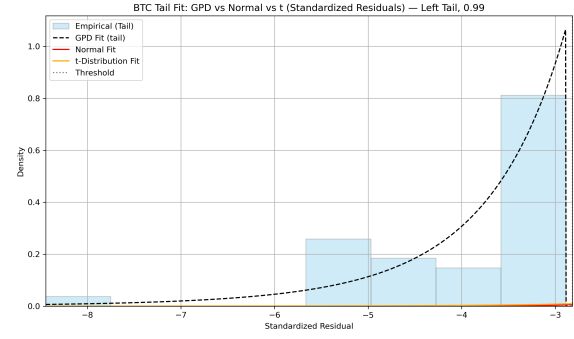
Since low-order GARCH models are the most widely used, this thesis uses GARCH(1,1) because it is relatively simple to implement, but quite effective. Therefore, to provide a more detailed analysis of the tails of these assets, EVT is combined with a GARCH(1,1) model, which helps eliminate autocorrelation between the logarithmic return series of each asset. In the book titled Quantitative Risk Management, there is a detailed chapter corresponding to EVT, where the authors mention: “To get round the problem that we do not observe data directly from the innovation distribution, we treat the residuals from the GARCH analysis as our data and apply the GPD tail estimation method of Section 5.2.3 to the residuals” (McNeil et al., 2015).

Following this logic, for each analysed asset, first a GARCH(1,1) model was fitted to the log returns and then the GPD, normal and t-distributions were fitted to the excess tail values with the aim of removing time-varying volatility and therefore allowing EVT to focus on the tail behaviour of the standardised residuals.

Comparison of the Results: When looking at the plots, for each asset the GPD seems to better approximate the extreme excess values than the t or normal distributions. Furthermore, at the 95% level, the GPD fit looks relatively accurate, but when moving to the 99% level, the fit becomes less precise (although still better than the estimate of the normal or t-distributions). The results also corroborate the findings from the previous section, where the behaviour of the left tails were compared.

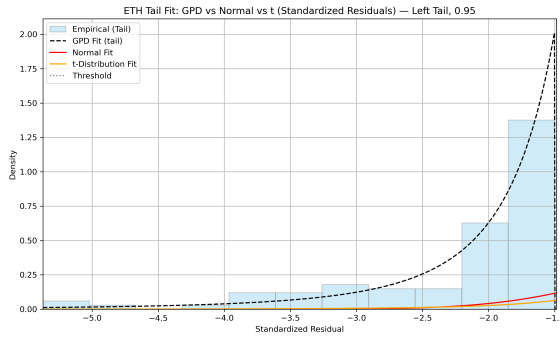


(a) 95% Quantile of the Distribution

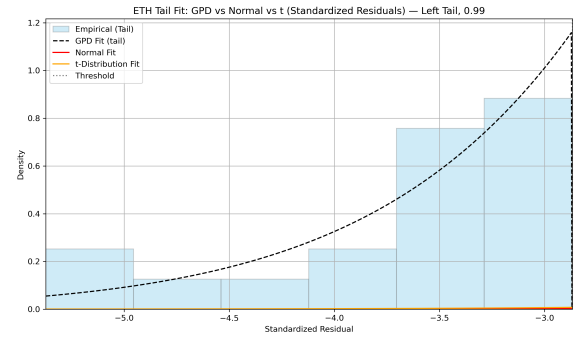


(b) 99% Quantile of the Distribution

Figure 28: Bitcoin GARCH Tail Fit

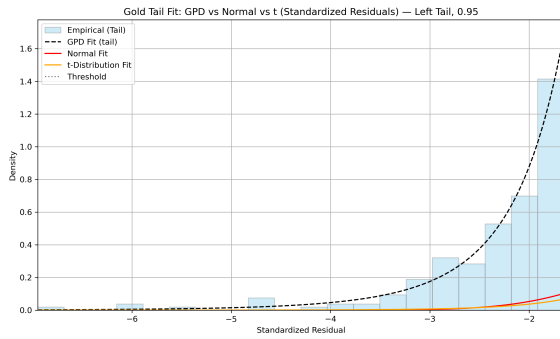


(a) 95% Quantile of the Distribution

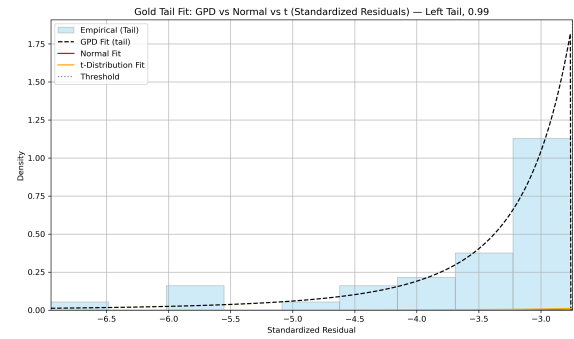


(b) 99% Quantile of the Distribution

Figure 29: Ethereum GARCH Tail Fit



(a) 95% Quantile of the Distribution



(b) 99% Quantile of the Distribution

Figure 30: Gold GARCH Tail Fit

4 Can EVT-based models provide better risk assessment for extreme events in the cryptocurrency market?

Previously in section 3.1, which explored the tail risk characteristics of cryptocurrencies and traditional financial assets, it was observed that EVT-based models appeared to fit the left tails of certain return distributions adequately. Based on this, the aim in this section is to conduct a risk assessment of different assets to check whether EVT-based models can indeed yield more accurate results when calculating specific risk metrics. Two of these metrics were examined in particular, the Value-at-Risk was calculated for different assets using the Gaussian Distribution, Student's t-distribution, and the Generalised Pareto Distribution. Furthermore, after performing the VaR calculations, the Conditional Value-at-Risk, more commonly known as Expected Shortfall, was also calculated. For each asset, the results were aggregated, and based on two selected frameworks, the best-performing model was chosen under the given assumptions.

4.1 Value-at-Risk

4.1.1 History and definition

Value at Risk (VaR) is one of the most commonly used risk metrics in the financial world and is also incorporated into the Basel Framework. For the corresponding calculations, the definition of VaR was used from the book on Quantitative Risk Management by Alexander J. McNeil, Rüdiger Frey and Paul Embrechts (McNeil et al., 2015).

To continue, Value at Risk was first introduced in the early 1990s and was formalised and widely disseminated by the RiskMetrics system of J.P. Morgan, which was released in October 1994 (J.P. Morgan & Reuters, 1996). Furthermore, the Group of Thirty (G30), an independent organisation comprising the world's most influential economists, also recommended the use of VaR for financial institutions in its 1993 study. They claimed that market risk is best measured through VaR and using probability analysis based on a common confidence interval and time horizon (Group of Thirty, Global Derivatives Study Group, 1993).

Consequently, VaR was added to the Basel I Accord Framework by the document titled: Amendment to Capital Accord to Incorporate Market Risks in January 1996. The primary reason for incorporating VaR into Basel I was that regulators stipulated that banks hold capital against the risk of losses, which was not adequately covered by the original version (Group of Thirty, Global Derivatives Study Group, 1993). The use of VaR via internal models was introduced as an expected method for banks to measure and calculate market risk capital changes (Basle Committee on Banking Supervision, 1996).

4.1.2 Comparison of VaR Models

After establishing the importance of VaR as a commonly used risk metric, it is also crucial to examine the methods available for comparing the accuracy of different VaR models. Consequently, a magnitude loss function was added to the backtesting process, which addresses the magnitude of the exceptions. Jose A. Lopez proposed this particular loss function in the 1998 FRBSF Economic Review (Lopez, 1998). It is also worth noting that the literature on evaluating the accuracy of VaR metrics is quite extensive, therefore, this particular evaluation method was chosen due to its relative simplicity compared to the other available models. Since the magnitude of the observed breaches can be quite large, Lopez incorporates this into

the loss function by introducing a magnitude term into the binomial loss function (Lopez, 1998). Therefore, this loss function represents both the total number of violations and their squared distances from the corresponding VaR (Zhang & Nadarajah, 2018).

$$L_t(x_t, \text{VaR}_t(\alpha)) = \begin{cases} 1 + [x_t - \text{VaR}_t(\alpha)]^2, & x_t \leq \text{VaR}_t \\ 0, & x_t > \text{VaR}_t \end{cases}$$

Given a data set on profits / losses, the sample average is

$$\hat{L} = \frac{1}{n} \sum_{t=1}^n L_t(x_t, \text{VaR}_t(\alpha)).$$

The distribution of $\hat{L}$ can be simulated and thus a rejection rule can be determined (Zhang & Nadarajah, 2018).

4.1.3 Formulas

To calculate the VaR for the Gaussian and t-distributions, the formulas presented in the QRM book can be utilised.

VaR for normal and t loss distributions: Suppose that the loss distribution F_L is normal with mean μ and variance σ^2 , and fix $\alpha \in (0, 1)$ (McNeil et al., 2015). This can be formally expressed as:

$$\text{VaR}_\alpha = \mu + \sigma \Phi^{-1}(\alpha), \quad (1)$$

We can model the tail risks with Extreme Value Theory, and thus calculate the VaR by utilising the Generalised Pareto Distribution. Using the expression from the Quantitative Risk Management book (McNeil et al., 2015), for $\alpha \geq F(u)$ we have that the GPD VaR is equal to:

$$\text{VaR}_\alpha = q_\alpha(F) = u + \frac{\beta}{\xi} \left(\left(\frac{1 - \alpha}{\bar{F}(u)} \right)^{-\xi} - 1 \right). \quad (2)$$

4.1.4 Backtesting Results

After selecting the formulas from the QRM Book (McNeil et al., 2015), for the three different VaR calculations, the backtesting process began using historical return data for Bitcoin and Ethereum. For Bitcoin, cleaned return data was used from 20 July 2010 to 16 May 2025, which yielded 3869 return values overall. For Ethereum, due to its later introduction, 1896 data points were obtainable from 9 February 2018 to 16 May 2025, but both datasets were large enough to carry out the backtesting successfully.

When executing the backtest, the initial step involved identifying the extreme losses. Several selection methodologies for extreme value thresholds were considered, however, given the limited sample size, the 5% lower tail of the return distribution was ultimately selected. Nevertheless, applying the backtest to the 1% tail would have been interesting, but this approach would have resulted in fewer than 25 extreme values in both the training and testing periods. Such a small number of extreme observations could potentially lead to unreliable parameter estimation and ambiguous backtesting results due to insufficient data for proper model fitting, thus the 5% lower tail was selected. To the selected extreme losses, the Gaussian, Student's t , and Generalised Pareto distributions were fitted to estimate the corresponding parameters, with which the calculation of the VaR estimates was possible.

Overall, the backtesting process itself consisted of multiple parts. To begin, the first few years of the time series were isolated and used solely for fitting the three different distributions

to obtain a stable model fit for the backtest and to calculate the initial three VaRs. After the fitting, the quarterly comparisons followed, where the estimated VaRs were compared to the actual losses that occurred in that quarter. As a basis for comparison, the empirical 5% VaR was also calculated for the quarter under analysis, to gauge the extent to which the three estimated models differed. For each loss within the quarter, it was examined whether any of the three VaR estimates were breached and, if so, by how much. This provided the foundation for the accuracy assessment, as VaR breaches—cases where realised losses surpass the estimated VaR—can expose both the risk manager and the institution to regulatory scrutiny and financial risk. Following the idea of the Lopez loss function, in case of a breach, the squared difference between the VaR and the loss were recorded and these values were averaged for the different VaRs for each quarter, and the VaR type that had the lowest average was deemed as the most accurate for the quarter, the amount of these “wins” were counted when deciding the best method for the overall backtest.

Another important feature of the backtest is that after the selection of the best method for the examined quarter, the backtesting loop moved on to the following quarter, but the previous “old” quarter was added to the refitting pool of returns for the models. In other words, after each quarter, the three VaR models were refitted using the return data that was isolated at the beginning, which was now extended with the “old” quarter. This step was included to incorporate “learning” and thus acquire better VaR estimates for the future quarters in the backtesting.

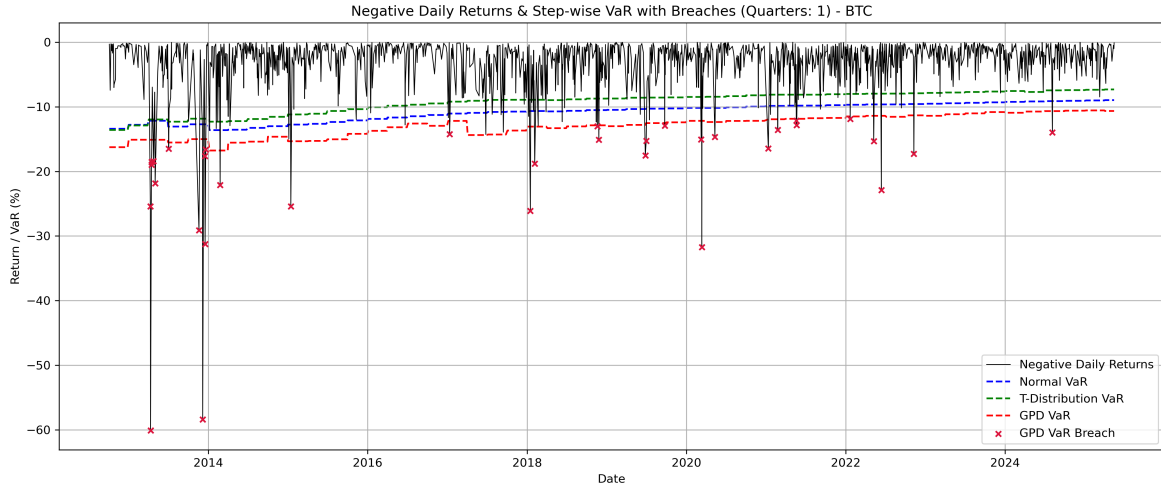


Figure 31: Bitcoin Quarterly VaR Comparison

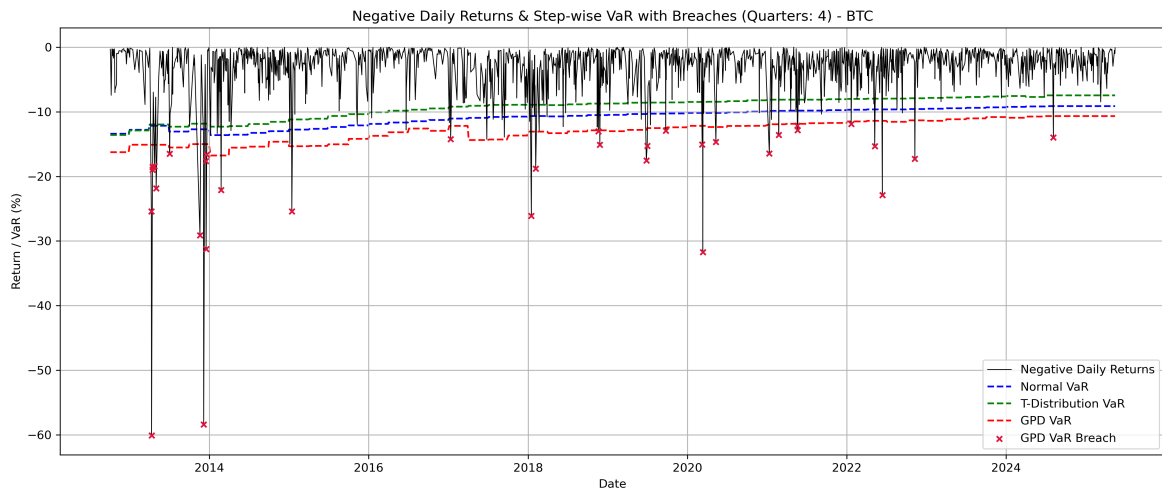


Figure 32: Bitcoin Yearly VaR Comparison

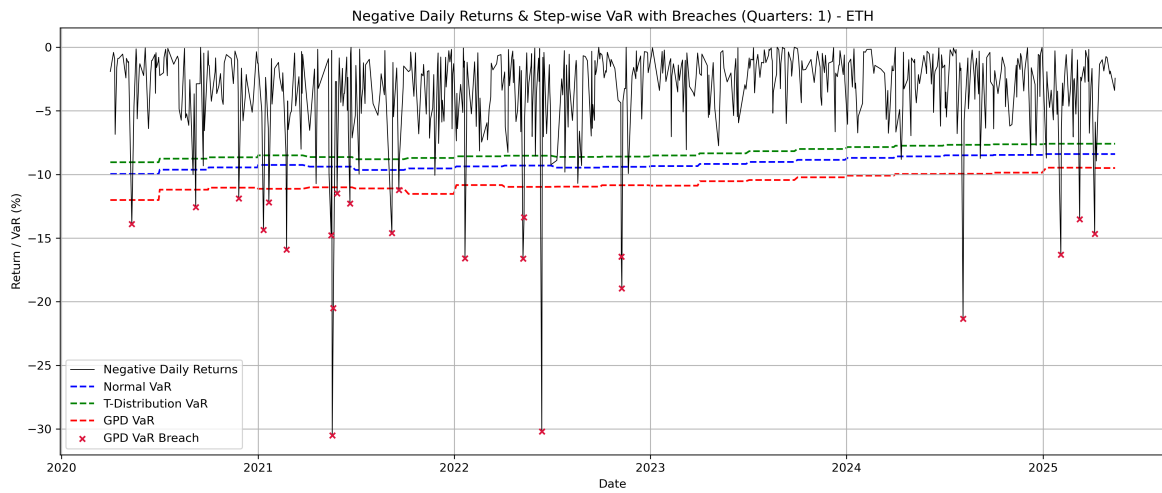


Figure 33: Ethereum Quarterly VaR Comparison

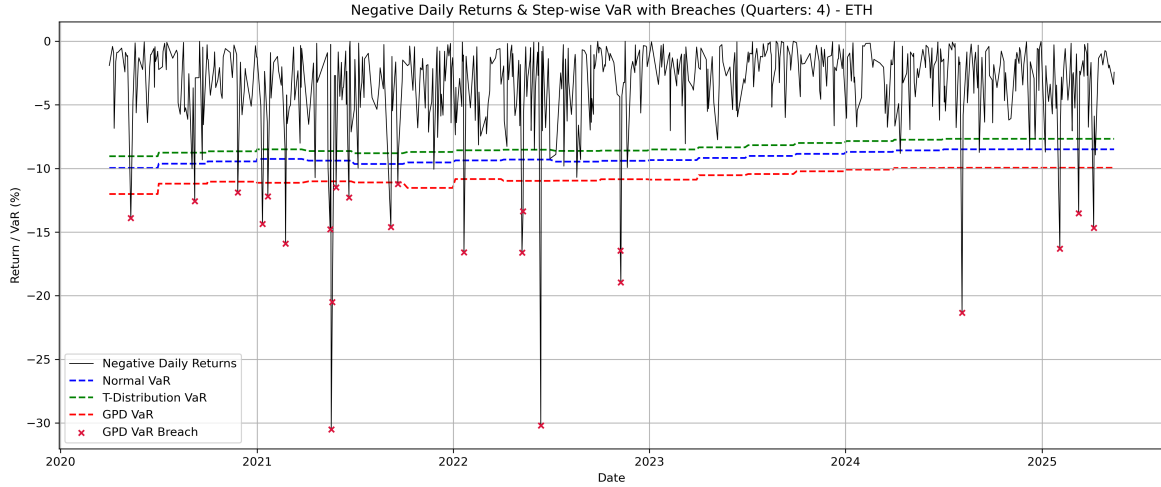


Figure 34: Ethereum Yearly VaR Comparison

The backtesting results indicate that the GPD-based VaR is the most accurate of the three VaR models tested, when evaluating both quarterly and yearly performance. When taking a look at Appendix Tables A, B, C, D; we see that in each case, the GPD VaR gets selected as the best VaR method the most times when comparing the Average Lopez Loss.

This is also corroborated by the VaR comparison plots, where the black line depicts the daily return, and the coloured lines display the forecasted VaR that is calculated on a quarterly/yearly basis. Moreover, the cases where the forecasted GPD VaR is breached are highlighted with red crosses. When examining both quarterly and yearly VaR comparisons for Bitcoin and Ethereum, it can be clearly observed that the GPD VaR gets breached by the black daily return line the least number of times.

Overall, the results suggest that GPD-based risk models can potentially provide better risk assessment for extreme events in the case of cryptocurrencies. In order to further test this hypothesis, it is crucial to examine the Expected Shortfall as well.

4.2 Expected Shortfall

4.2.1 History and definition

Similarly to the VaR, the Expected Shortfall, also known as Conditional Value at Risk, appeared around the late 1990s. One of the first papers to mention Expected Shortfall as a potentially appropriate risk measure was published in 1999 with the title: “Coherent Measures of Risk”. In this journal article, the authors showed that VaR has significant limitations compared to ES, and ES-based risk measures can be more robust and coherent. They argue that while VaR is widely used, its mathematical properties (especially the lack of subadditivity) make it an unreliable measure for risk aggregation, concentration, and allocation (Artzner et al., 1999).

The superior tail properties of Expected Shortfall gained significant attention after the 2008 Financial Crisis. Moreover, the Basel Committee on Banking Supervision also shifted its focus from VaR to Expected Shortfall. Under the Internal Models Approach (IMA) in their 2019 document titled “Minimum Capital Requirements for Market Risk”, the Basel Committee mandates the use of Expected Shortfall for capital calculation. In the MAR33.2 section of the document, it is stated that: “ES must be computed on a daily basis for the bank-wide internal models to determine market risk capital requirements. ES must also be

computed on a daily basis for each trading desk that uses the internal models approach (IMA)” (Basle Committee on Banking Supervision, 2019).

4.2.2 Comparison of ES Models

4.2.3 Formulas

Thanks to the Quantitative Risk Management book (McNeil et al., 2015), we have explicit expressions for the calculation of different types of expected shortfalls, the proposed formulas are the following:

Expected Shortfall for Gaussian Loss Distribution: Suppose that the loss distribution F_L is normal with mean μ and variance σ^2 ; fix $\alpha \in (0, 1)$ (McNeil et al., 2015). Formally expressed as:

$$\text{ES}_\alpha = \mu + \sigma \frac{\varphi(\Phi^{-1}(\alpha))}{1 - \alpha}, \quad (3)$$

Expected Shortfall for the Student t Loss Distribution: Suppose the loss L is such that $\tilde{L} = (L - \mu)/\sigma$ has a standard t distribution with ν degrees of freedom and suppose further that $\nu > 1$ (McNeil et al., 2015). The ES of the standard t distribution can be calculated by direct integration to be the following, where t_ν denotes the degree of freedom and g_ν the density of standard t :

$$\text{ES}_\alpha(\tilde{L}) = \frac{g_\nu(t_\nu^{-1}(\alpha))}{1 - \alpha} \left(\frac{\nu + (t_\nu^{-1}(\alpha))^2}{\nu - 1} \right), \quad (4)$$

Expected Shortfall for the Generalized Pareto Distribution: Assuming that $\xi < 1$, the associated expected shortfall can be calculated and we obtain the following expression (McNeil et al., 2015):

$$\text{ES}_\alpha = \frac{1}{1 - \alpha} \int_\alpha^1 q_x(F) dx = \frac{\text{VaR}_\alpha}{1 - \xi} + \frac{\beta - \xi u}{1 - \xi}. \quad (5)$$

4.2.4 Backtesting Results

We have previously seen that there are several ways to assess the accuracy of different VaR models, and accordingly, there are many available options for ES models as well. For example, in the 2014 paper by Acerbi & Szekely, three different ways are highlighted for backtesting ES models. It is possible to test the ES after VaR by considering the magnitude of realised exceptions relative to the model predictions, however this method is insensitive to a large amount of exceptions (Acerbi & Szekely, 2014). It is also possible to test ES directly and look at both the frequency and magnitude of tail events (Acerbi & Szekely, 2014). Lastly, it is also possible to estimate ES from realised ranks by estimating it from the entire vector of realised ranks and then averaging the results over the entire period (Acerbi & Szekely, 2014). Although this last test is less intuitive than the previous two, it is highly versatile and can be adapted to other statistical applications.

For the Conditional VaR, the same backtesting loop was utilised that was used for the simple VaR, and thus the steps and the logic remained the same. Instead of VaR, the Expected Shortfall calculation was performed, and the losses were compared to the three predicted ES values for each examined quarter. For the method to assess the accuracy of the different ES models, the above-mentioned paper provided inspiration. First, the forecasted VaRs were compared to the actual losses in the examined quarter, and the instances where the realised losses exceeded the VaR predictions were recorded. Where a breach was detected, the difference between the loss and the forecasted Expected Shortfall was calculated to see how

accurately the model predicted the actual loss. As the final step, the average of these excess losses for each ES model was calculated, and the one with the lowest average excess loss was selected as the most accurate for the examined period. These quarterly "wins" were counted for each ES method at the end of the backtest, and the most accurate one was selected as the overall best.

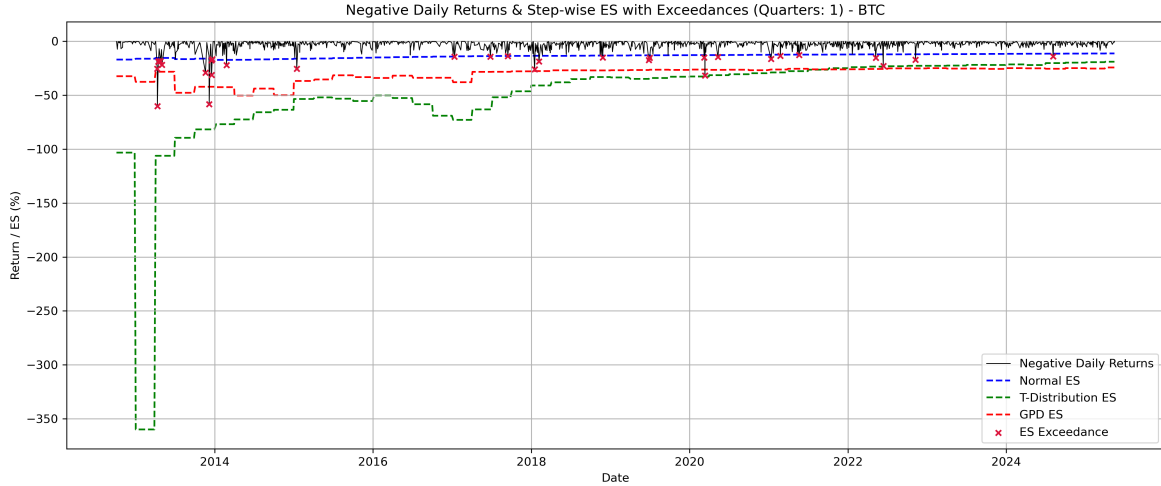


Figure 35: Bitcoin Quarterly ES Comparison

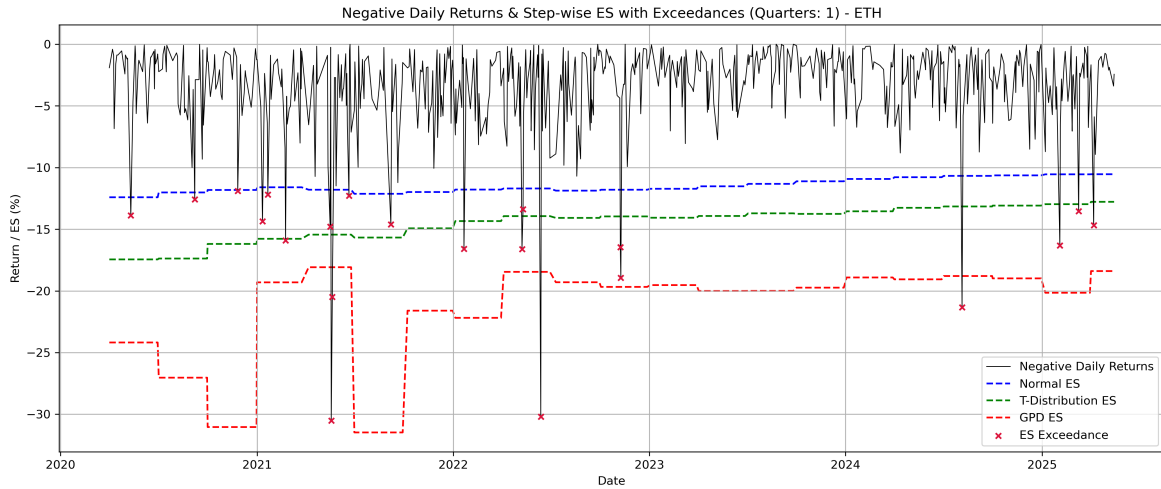


Figure 36: Ethereum Quarterly ES Comparison

For both cryptocurrencies, using the above-mentioned method for comparison, we get that the GPD based Expected Shortfall is the most accurate from the three, when comparing the ES forecasts using a rolling quarterly window. This is shown in Appendix Table E, where the GPD ES gets selected as the most accurate ES method for Ethereum, and it is the same for Bitcoin when taking a look at Appendix Table F.

The Expected Shortfall plots also confirm the accuracy of the GPD ES. The black line depicts the daily returns for Bitcoin and Ethereum, and the coloured lines represent the different forecasted Expected Shortfall values for the given quarter. It is important to note that the step-like shape of the ES lines are due to the fact that we calculate the ES forecasts

on a quarterly basis, whereas the returns (black colour) are of daily frequency. Furthermore, the instances where a given daily loss was larger than the Expected Shortfall forecast, are flagged with a red cross, and these scenarios are the ones that are considered "breaches". It can be seen that the forecasted GPD ES is breached the least amount of times in the case of both Ethereum and Bitcoin.

It is also important to highlight that for the Bitcoin quarter 31 December 2012 – 31 March 2013, the t-distribution based forecast was -359,89% which was considerably lower than the others and it further demonstrates the inadequate fitting of the t-distribution-based ES.

All in all, under our assumptions, using the selected return data and this type of expanding training window backtesting, we can conclude that EVT-based models can provide better risk assessment for extreme return events in the cryptocurrency market.

5 Conclusion

This thesis aimed to compare the risk and price behavior of cryptocurrencies with traditional assets using clear and practical statistical methods. The results confirm that cryptocurrencies experience much higher volatility, more frequent extremes, and stronger clustering of large price moves than assets like stocks or gold. It was also demonstrated that standard risk models often underestimate the true risks in these markets, while models focused on extreme values are more effective at identifying and measuring rare, significant price swings. Overall, these findings suggest that investors and risk managers need to use specialized tools to properly assess and manage risk in cryptocurrency markets.

Looking ahead, the unique risk characteristics of cryptocurrencies make them both an exciting opportunity and a considerable challenge for market participants. As these markets continue to mature, it will be important to keep updating risk management approaches and to consider new forms of analysis, including higher-frequency data and more advanced modeling techniques. While this thesis uses straightforward methods and publicly available data, future research could further explore these topics with more detailed datasets and a wider range of cryptocurrencies.

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6 Appendix

A ETH VaR Results – Quarterly

Empirical VaR (Test)	Normal VaR	T- Distribution VaR	GPD VaR	Average Lopez Loss (Normal)	Average Lopez Loss (t)	Average Lopez Loss (GPD)	Best VaR Method	Start	End
-5.3876%	-9.9602%	-9.0400%	-12.0079%	1.5175%	1.5187%	1.5157%	GPD VaR	2020-03-31	2020-06-30
-6.9753%	-9.6253%	-8.7597%	-11.1952%	2.9864%	4.4801%	1.4928%	GPD VaR	2020-06-30	2020-09-30
-4.4211%	-9.4477%	-8.6522%	-11.0376%	1.4934%	1.4941%	1.4926%	GPD VaR	2020-09-30	2020-12-31
-8.3330%	-9.2528%	-8.4989%	-11.1260%	4.6275%	4.6312%	4.6207%	GPD VaR	2020-12-31	2021-03-31
-12.0799%	-9.3861%	-8.6326%	-11.0110%	10.6990%	10.7097%	7.6494%	GPD VaR	2021-03-31	2021-06-30
-8.0635%	-9.6476%	-8.8032%	-11.0993%	4.4817%	4.4837%	2.9869%	GPD VaR	2021-06-30	2021-09-30
-6.6917%	-9.5283%	-8.7039%	-11.5284%	1.4926%	1.4928%	0.0000%	GPD VaR	2021-09-30	2021-12-31
-7.3457%	-9.3682%	-8.5737%	-10.8336%	1.5465%	1.5483%	1.5435%	GPD VaR	2021-12-31	2022-03-31
-9.0670%	-9.3010%	-8.5256%	-10.9757%	4.6221%	7.6604%	4.6070%	GPD VaR	2022-03-31	2022-06-30
-9.1335%	-9.4632%	-8.6222%	-10.9574%	2.9853%	7.4637%	0.0000%	GPD VaR	2022-06-30	2022-09-30
-5.3257%	-9.3997%	-8.5963%	-10.8469%	4.5668%	4.5713%	3.0450%	GPD VaR	2022-09-30	2022-12-31
-3.7947%	-9.3410%	-8.5077%	-10.8788%	0.0000%	0.0000%	0.0000%	-	2022-12-31	2023-03-31
-5.4409%	-9.1746%	-8.3408%	-10.5300%	0.0000%	0.0000%	0.0000%	-	2023-03-31	2023-06-30
-3.4589%	-9.0166%	-8.1709%	-10.4367%	0.0000%	0.0000%	0.0000%	-	2023-06-30	2023-09-30
-3.7944%	-8.8528%	-7.9993%	-10.2231%	0.0000%	0.0000%	0.0000%	-	2023-09-30	2023-12-31
-5.6260%	-8.6986%	-7.8468%	-10.0918%	0.0000%	0.0000%	0.0000%	-	2023-12-31	2024-03-31
-5.9909%	-8.5808%	-7.7422%	-9.9620%	1.5385%	1.5386%	0.0000%	GPD VaR	2024-03-31	2024-06-30
-6.2196%	-8.4952%	-7.6711%	-9.9474%	3.0553%	3.0587%	1.5348%	GPD VaR	2024-06-30	2024-09-30
-6.2940%	-8.4664%	-7.6298%	-9.8566%	1.4925%	2.9852%	0.0000%	GPD VaR	2024-09-30	2024-12-31
-6.9906%	-8.3961%	-7.5864%	-9.4678%	4.6291%	4.6327%	3.0867%	GPD VaR	2024-12-31	2025-03-31
-6.8018%	-8.3969%	-7.5874%	-9.4943%	5.7256%	5.7291%	2.8648%	GPD VaR	2025-03-31	2025-06-30

B ETH VaR Results - Yearly

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Empirical VaR (Test)	Normal VaR	T- Distribution VaR	GPD VaR	Average Lopez Loss (Normal)	Average Lopez Loss (t)	Average Lopez Loss (GPD)	Best VaR Method	Start	End
-6.4749%	-9.9602%	-9.0400%	-12.0079%	2.6750%	3.0584%	1.9093%	GPD VaR	2020-03-31	2021-03-31
-9.2771%	-9.6253%	-8.7597%	-11.1952%	4.9876%	5.3737%	3.8362%	GPD VaR	2020-06-30	2021-06-30
-9.8238%	-9.4477%	-8.6522%	-11.0376%	5.3709%	5.7570%	4.2188%	GPD VaR	2020-09-30	2021-09-30
-9.8238%	-9.2528%	-8.4989%	-11.1260%	5.3717%	5.7576%	3.8368%	GPD VaR	2020-12-31	2021-12-31
-8.1217%	-9.3861%	-8.6326%	-11.0110%	4.6067%	4.9921%	3.0737%	GPD VaR	2021-03-31	2022-03-31
-7.5566%	-9.6476%	-8.8032%	-11.0993%	3.0748%	3.4596%	2.3069%	GPD VaR	2021-06-30	2022-06-30
-8.1217%	-9.5283%	-8.7039%	-11.5284%	2.6925%	3.8403%	1.5421%	GPD VaR	2021-09-30	2022-09-30
-8.3529%	-9.3682%	-8.5737%	-10.8336%	3.4750%	4.6282%	2.3197%	GPD VaR	2021-12-31	2022-12-31
-8.2584%	-9.3010%	-8.5256%	-10.9757%	3.4600%	4.6084%	1.9275%	GPD VaR	2022-03-31	2023-03-31
-6.5836%	-9.4632%	-8.6222%	-10.9574%	1.9138%	3.0602%	0.7669%	GPD VaR	2022-06-30	2023-06-30
-5.1819%	-9.3997%	-8.5963%	-10.8469%	1.1548%	1.1560%	0.7700%	GPD VaR	2022-09-30	2023-09-30
-4.5823%	-9.3410%	-8.5077%	-10.8788%	0.0000%	0.0000%	0.0000%	-	2022-12-31	2023-12-31
-5.1819%	-9.1746%	-8.3408%	-10.5300%	0.0000%	0.0000%	0.0000%	-	2023-03-31	2024-03-31
-4.8992%	-9.0166%	-8.1709%	-10.4367%	0.0000%	0.3832%	0.0000%	Normal VaR and GPD VaR	2023-06-30	2024-06-30
-5.6732%	-8.8528%	-7.9993%	-10.2231%	0.3891%	1.1563%	0.3879%	GPD VaR	2023-09-30	2024-09-30
-6.2052%	-8.6986%	-7.8468%	-10.0918%	1.1511%	1.5337%	0.3865%	GPD VaR	2023-12-31	2024-12-31
-6.5515%	-8.5808%	-7.7422%	-9.9620%	2.3083%	2.6933%	1.1564%	GPD VaR	2024-03-31	2025-03-31
-6.6641%	-8.4952%	-7.6711%	-9.9474%	3.4908%	3.9282%	1.7480%	GPD VaR	2024-06-30	2025-06-30

C BTC VaR Results - Quarterly

Empirical VaR (Test)	Normal VaR	T- Distribution VaR	GPD VaR	Average Lopez Loss (Normal)	Average Lopez Loss (t)	Average Lopez Loss (GPD)	Best VaR Method	Start	End
Continued on next page(s)									
-4.9939%	-13.4000%	-13.5874%	-16.2520%	0.0000%	0.0000%	0.0000%	-	2012-09-30	2012-12-31
-2.8172%	-12.7830%	-12.8907%	-15.0993%	0.0000%	0.0000%	0.0000%	-	2012-12-31	2013-03-31
-18.8777%	-12.1613%	-11.9623%	-15.1059%	9.6450%	9.6505%	9.5712%	GPD VaR	2013-03-31	2013-06-30
-8.1002%	-13.0602%	-12.3057%	-15.5278%	1.5170%	1.5178%	1.5153%	GPD VaR	2013-06-30	2013-09-30
-17.3191%	-12.7104%	-11.8418%	-15.0037%	9.3639%	10.8804%	7.8140%	GPD VaR	2013-09-30	2013-12-31
-11.5812%	-13.6103%	-12.3040%	-16.7661%	3.0880%	4.6306%	1.5429%	GPD VaR	2013-12-31	2014-03-31
-6.7997%	-13.5506%	-12.2177%	-15.5507%	0.0000%	1.5152%	0.0000%	Normal VaR and GPD VaR	2014-03-31	2014-06-30
-5.5436%	-13.2633%	-11.8934%	-15.3933%	0.0000%	0.0000%	0.0000%	-	2014-06-30	2014-09-30
-5.0539%	-13.0030%	-11.5525%	-14.6344%	0.0000%	0.0000%	0.0000%	-	2014-09-30	2014-12-31
-10.0547%	-12.7536%	-11.1864%	-15.3333%	4.6410%	4.6497%	1.5541%	GPD VaR	2014-12-31	2015-03-31
-3.3600%	-12.6367%	-11.0725%	-15.2814%	0.0000%	0.0000%	0.0000%	-	2015-03-31	2015-06-30
-4.2022%	-12.3501%	-10.6482%	-15.0331%	0.0000%	0.0000%	0.0000%	-	2015-06-30	2015-09-30
-7.6212%	-12.1086%	-10.3680%	-14.1866%	0.0000%	1.4929%	0.0000%	Normal VaR and GPD VaR	2015-09-30	2015-12-31
-3.7416%	-11.8729%	-10.0660%	-13.7205%	0.0000%	1.5153%	0.0000%	Normal VaR and GPD VaR	2015-12-31	2016-03-31
-2.4887%	-11.6608%	-9.8297%	-13.1684%	1.5154%	1.5165%	0.0000%	GPD VaR	2016-03-31	2016-06-30
-2.8908%	-11.4646%	-9.6743%	-12.5931%	0.0000%	0.0000%	0.0000%	-	2016-06-30	2016-09-30

-1.1229%	-11.2706%	-9.4850%	-12.9466%	0.0000%	0.0000%	0.0000%	-	2016-09-30	2016-12-31
-8.5083%	-11.0475%	-9.2040%	-12.1841%	3.0785%	3.0818%	1.5391%	GPD VaR	2016-12-31	2017-03-31
-7.6753%	-10.9403%	-9.0704%	-14.3735%	1.5168%	1.5193%	0.0000%	GPD VaR	2017-03-31	2017-06-30
-8.0959%	-10.7989%	-8.9322%	-14.2906%	4.5470%	4.5508%	0.0000%	GPD VaR	2017-06-30	2017-09-30
-7.2946%	-10.7448%	-8.9145%	-13.6777%	0.0000%	1.5385%	0.0000%	Normal VaR and GPD VaR	2017-09-30	2017-12-31
-11.3452%	-10.6554%	-8.9137%	-13.0671%	7.7401%	12.3726%	3.1081%	GPD VaR	2017-12-31	2018-03-31
-7.3563%	-10.7160%	-8.9682%	-13.3105%	1.5389%	3.0789%	0.0000%	GPD VaR	2018-03-31	2018-06-30
-5.7695%	-10.6227%	-8.8470%	-13.0348%	0.0000%	0.0000%	0.0000%	-	2018-06-30	2018-09-30
-9.1188%	-10.5142%	-8.7319%	-12.8699%	4.5501%	6.0717%	3.0311%	GPD VaR	2018-09-30	2018-12-31
-3.4933%	-10.4752%	-8.7200%	-13.0052%	0.0000%	1.5386%	0.0000%	Normal VaR and GPD VaR	2018-12-31	2019-03-31
-7.6769%	-10.3484%	-8.5951%	-12.8024%	3.0850%	4.6287%	1.5419%	GPD VaR	2019-03-31	2019-06-30
-8.8563%	-10.2911%	-8.5742%	-12.5222%	4.5507%	6.0722%	3.0315%	GPD VaR	2019-06-30	2019-09-30
-4.7679%	-10.2562%	-8.5478%	-12.4061%	0.0000%	0.0000%	0.0000%	-	2019-09-30	2019-12-31
-7.0147%	-10.1688%	-8.4760%	-12.1772%	4.6195%	4.6344%	3.0895%	GPD VaR	2019-12-31	2020-03-31
-5.1681%	-10.1681%	-8.4458%	-12.3583%	1.5182%	1.5210%	1.5159%	GPD VaR	2020-03-31	2020-06-30
-4.2662%	-10.0774%	-8.3540%	-12.1909%	0.0000%	0.0000%	0.0000%	-	2020-06-30	2020-09-30
-1.9234%	-9.9753%	-8.2239%	-12.1745%	1.4925%	1.4931%	0.0000%	GPD VaR	2020-09-30	2020-12-31
-6.7980%	-9.8635%	-8.1232%	-11.9304%	4.6245%	4.6321%	3.0805%	GPD VaR	2020-12-31	2021-03-31
-9.6882%	-9.8214%	-8.1214%	-11.8616%	4.5477%	10.6132%	3.0305%	GPD VaR	2021-03-31	2021-06-30
-4.9771%	-9.8139%	-8.1262%	-11.7682%	1.4926%	1.4933%	0.0000%	GPD VaR	2021-06-30	2021-09-30
-5.7603%	-9.7462%	-8.0557%	-11.7264%	0.0000%	1.4926%	0.0000%	Normal VaR and GPD VaR	2021-09-30	2021-12-31
-6.5726%	-9.6856%	-8.0099%	-11.4947%	1.5392%	1.5408%	1.5385%	GPD VaR	2021-12-31	2022-03-31
-8.4489%	-9.6329%	-7.9690%	-11.3963%	3.0618%	6.1026%	3.0526%	GPD VaR	2022-03-31	2022-06-30

-6.4121%	-9.6357%	-7.9576%	-11.5510%	1.4926%	2.9862%	0.0000%	GPD VaR	2022-06-30	2022-09-30
-5.3795%	-9.5749%	-7.9074%	-11.3270%	3.0393%	3.0443%	1.5205%	GPD VaR	2022-09-30	2022-12-31
-3.1928%	-9.5230%	-7.8494%	-11.3945%	0.0000%	1.5385%	0.0000%	Normal VaR and GPD VaR	2022-12-31	2023-03-31
-3.8582%	-9.4542%	-7.7900%	-11.1564%	0.0000%	0.0000%	0.0000%	-	2023-03-31	2023-06-30
-3.6730%	-9.3849%	-7.7102%	-11.0198%	0.0000%	0.0000%	0.0000%	-	2023-06-30	2023-09-30
-2.5388%	-9.3136%	-7.6402%	-10.8167%	0.0000%	1.5385%	0.0000%	Normal VaR and GPD VaR	2023-09-30	2023-12-31
-4.8557%	-9.2366%	-7.5584%	-10.9244%	0.0000%	0.0000%	0.0000%	-	2023-12-31	2024-03-31
-4.9895%	-9.1743%	-7.7067%	-10.7367%	0.0000%	0.0000%	0.0000%	-	2024-03-31	2024-06-30
-4.0807%	-9.1228%	-7.4576%	-10.6671%	1.5187%	1.5216%	1.5168%	GPD VaR	2024-06-30	2024-09-30
-3.9445%	-9.0757%	-7.4192%	-10.6017%	0.0000%	0.0000%	0.0000%	-	2024-09-30	2024-12-31
-5.1642%	-9.0167%	-7.3663%	-10.5630%	0.0000%	1.5387%	0.0000%	Normal VaR and GPD VaR	2024-12-31	2025-03-31
-3.9948%	-8.9674%	-7.3144%	-10.6483%	0.0000%	0.0000%	0.0000%	-	2025-03-31	2025-06-30

D BTC VaR Results - Yearly

	Empirical VaR (Test)	Normal VaR	T- Distribution VaR	GPD VaR	Average Lopez Loss (Normal)	Average Lopez Loss (t)	Average Lopez Loss (GPD)	Best VaR Method	Start	End
	Continued on next page(s)									
52	-7.4411%	-13.4000%	-13.5874%	-16.2520%	2.7773%	2.7761%	2.7607%	GPD VaR	2012-09-30	2013-09-30
	-12.2152%	-12.7830%	-12.8907%	-15.0993%	5.1649%	5.1635%	4.7540%	GPD VaR	2012-12-31	2013-12-31
	-13.6643%	-12.1613%	-11.9623%	-15.1059%	7.1131%	7.1162%	5.1572%	GPD VaR	2013-03-31	2014-03-31
	-11.4962%	-13.0602%	-12.3057%	-15.5278%	3.9376%	4.3268%	2.7708%	GPD VaR	2013-06-30	2014-06-30
	-10.3923%	-12.7104%	-11.8418%	-15.0037%	3.9247%	4.6951%	2.3818%	GPD VaR	2013-09-30	2014-09-30
	-6.5881%	-13.6103%	-12.3040%	-16.7661%	0.7661%	1.5305%	0.3828%	GPD VaR	2013-12-31	2014-12-31
	-6.8279%	-13.5506%	-12.2177%	-15.5507%	1.1505%	1.5338%	0.3854%	GPD VaR	2014-03-31	2015-03-31
	-5.5457%	-13.2633%	-11.8934%	-15.3933%	1.1508%	1.1525%	0.3855%	GPD VaR	2014-06-30	2015-06-30
	-5.1876%	-13.0030%	-11.5525%	-14.6344%	1.1511%	1.1530%	0.7678%	GPD VaR	2014-09-30	2015-09-30
	-6.0998%	-12.7536%	-11.1864%	-15.3333%	1.1514%	1.5353%	0.3856%	GPD VaR	2014-12-31	2015-12-31
	-4.2620%	-12.6367%	-11.0725%	-15.2814%	0.0000%	0.3803%	0.0000%	Normal VaR and GPD VaR	2015-03-31	2016-03-31
	-4.2762%	-12.3501%	-10.6482%	-15.0331%	0.3802%	1.1409%	0.0000%	GPD VaR	2015-06-30	2016-06-30
	-4.1341%	-12.1086%	-10.3680%	-14.1866%	0.3802%	1.1410%	0.0000%	GPD VaR	2015-09-30	2016-09-30
	-3.1527%	-11.8729%	-10.0660%	-13.7205%	0.3817%	0.7637%	0.0000%	GPD VaR	2015-12-31	2016-12-31
	-6.4522%	-11.6608%	-9.8297%	-13.1684%	1.1453%	1.1462%	0.3817%	GPD VaR	2016-03-31	2017-03-31
	-7.7620%	-11.4646%	-9.6743%	-12.5931%	1.1456%	1.1468%	0.7636%	GPD VaR	2016-06-30	2017-06-30

Ξ	-7.9257%	-11.2706%	-9.4850%	-12.9466%	1.5335%	2.3018%	1.1496%	GPD VaR	2016-09-30	2017-09-30
	-8.1457%	-11.0475%	-9.2040%	-12.1841%	2.3088%	2.6957%	1.1543%	GPD VaR	2016-12-31	2017-12-31
	-8.8201%	-10.9403%	-9.0704%	-14.3735%	3.0772%	4.9989%	0.7723%	GPD VaR	2017-03-31	2018-03-31
	-9.1172%	-10.7989%	-8.9322%	-14.2906%	3.0772%	5.3820%	0.7724%	GPD VaR	2017-06-30	2018-06-30
	-8.5949%	-10.7448%	-8.9145%	-13.6777%	2.3196%	4.2475%	0.7762%	GPD VaR	2017-09-30	2018-09-30
	-9.1172%	-10.6554%	-8.9137%	-13.0671%	3.4614%	5.3833%	1.1574%	GPD VaR	2017-12-31	2018-12-31
	-6.5570%	-10.7160%	-8.9682%	-13.3105%	1.5396%	2.6954%	0.3847%	GPD VaR	2018-03-31	2019-03-31
	-6.5764%	-10.6227%	-8.8470%	-13.0348%	1.9260%	3.0827%	0.7702%	GPD VaR	2018-06-30	2019-06-30
	-7.5117%	-10.5142%	-8.7319%	-12.8699%	3.0694%	4.6065%	1.9170%	GPD VaR	2018-09-30	2019-09-30
	-6.2283%	-10.4752%	-8.7200%	-13.0052%	1.9115%	3.0594%	0.7643%	GPD VaR	2018-12-31	2019-12-31
	-7.0235%	-10.3484%	-8.5951%	-12.8024%	2.6933%	3.8451%	1.9234%	GPD VaR	2019-03-31	2020-03-31
	-6.0293%	-10.2911%	-8.5742%	-12.5222%	2.3105%	3.0799%	1.9232%	GPD VaR	2019-06-30	2020-06-30
	-5.4171%	-10.2562%	-8.5478%	-12.4061%	1.1598%	1.5445%	1.1553%	GPD VaR	2019-09-30	2020-09-30
	-5.3122%	-10.1688%	-8.4760%	-12.1772%	1.5403%	1.9250%	1.1558%	GPD VaR	2019-12-31	2020-12-31
	-5.3135%	-10.1681%	-8.4458%	-12.3583%	1.5295%	1.9137%	1.1459%	GPD VaR	2020-03-31	2021-03-31
	-6.8206%	-10.0774%	-8.3540%	-12.1909%	2.2926%	4.2040%	1.1458%	GPD VaR	2020-06-30	2021-06-30
	-7.6132%	-9.9753%	-8.2239%	-12.1745%	2.6744%	4.5862%	1.1458%	GPD VaR	2020-09-30	2021-09-30
	-7.6132%	-9.8635%	-8.1232%	-11.9304%	2.6746%	4.5863%	1.5276%	GPD VaR	2020-12-31	2021-12-31
	-7.5555%	-9.8214%	-8.1214%	-11.8616%	1.9091%	3.8193%	1.1451%	GPD VaR	2021-03-31	2022-03-31
	-6.9520%	-9.8139%	-8.1262%	-11.7682%	1.5346%	2.6828%	1.1502%	GPD VaR	2021-06-30	2022-06-30
	-6.9520%	-9.7462%	-8.0557%	-11.7264%	1.5347%	3.0647%	1.1503%	GPD VaR	2021-09-30	2022-09-30
	-7.0594%	-9.6856%	-8.0099%	-11.4947%	2.3091%	3.4631%	1.5394%	GPD VaR	2021-12-31	2022-12-31
	-6.2257%	-9.6329%	-7.9690%	-11.3963%	1.9186%	3.4495%	1.1520%	GPD VaR	2022-03-31	2023-03-31

-5.1276%	-9.6357%	-7.9576%	-11.5510%	1.1473%	1.9122%	0.3829%	GPD VaR	2022-06-30	2023-06-30
-3.9312%	-9.5749%	-7.9074%	-11.3270%	0.7686%	1.1530%	0.3845%	GPD VaR	2022-09-30	2023-09-30
-3.6490%	-9.5230%	-7.8494%	-11.3945%	0.0000%	0.7692%	0.0000%	Normal VaR and GPD VaR	2022-12-31	2023-12-31
-4.0748%	-9.4542%	-7.7900%	-11.1564%	0.0000%	0.3831%	0.0000%	Normal VaR and GPD VaR	2023-03-31	2024-03-31
-4.5715%	-9.3849%	-7.7102%	-11.0198%	0.0000%	0.3831%	0.0000%	Normal VaR and GPD VaR	2023-06-30	2024-06-30
-4.3969%	-9.3136%	-7.6402%	-10.8167%	0.3840%	0.7678%	0.3835%	GPD VaR	2023-09-30	2024-09-30
-4.7940%	-9.2366%	-7.5584%	-10.9244%	0.3825%	0.7649%	0.3820%	GPD VaR	2023-12-31	2024-12-31
-4.8883%	-9.1743%	-7.7067%	-10.7367%	0.3840%	0.7678%	0.3835%	GPD VaR	2024-03-31	2025-03-31
-4.2204%	-9.1228%	-7.4576%	-10.6671%	0.4358%	0.8715%	0.4353%	GPD VaR	2024-06-30	2025-06-30

E ETH ES Results - Quarterly

	Empirical ES (Test)	Normal ES	T- Distribution ES	GPD ES	Mean Excess (Normal)	Mean Excess (t)	AMean Excess (GPD)	Best ES Method	Start	End
	Continued on next page(s)									
55	-8.1903%	-12.4092%	-17.4468%	-24.1807%	1.4782%	-3.5594%	-10.2933%	GPD	2020-03-31	2020-06-30
	-9.7620%	-12.0188%	-17.3752%	-27.0330%	-0.7320%	-6.7431%	-14.4626%	GPD	2020-06-30	2020-09-30
	-6.8200%	-11.8179%	-16.1921%	-31.0356%	0.0699%	-4.3043%	-19.1477%	GPD	2020-09-30	2020-12-31
	-12.7153%	-11.6006%	-15.7759%	-19.3065%	2.5489%	-1.6264%	-5.1570%	GPD	2020-12-31	2021-03-31
	-19.5140%	-11.7979%	-15.4383%	-18.0802%	3.9378%	0.2974%	-0.1711%	GPD	2021-03-31	2021-06-30
	-11.1381%	-12.1279%	-15.6742%	-31.4728%	-0.1967%	-3.7430%	-18.5620%	GPD	2021-06-30	2021-09-30
	-7.8958%	-11.9839%	-14.9262%	-21.6013%	-1.9205%	-4.8628%	No Breach	GPD	2021-09-30	2021-12-31
	-9.8868%	-11.7862%	-14.3403%	-22.1837%	4.7928%	2.2387%	-5.6046%	GPD	2021-12-31	2022-03-31
	-17.3498%	-11.6971%	-13.9380%	-18.4561%	8.3574%	1.6539%	1.5984%	GPD	2022-03-31	2022-06-30
	-9.7636%	-11.8723%	-14.0815%	-19.2980%	-1.6140%	-4.4916%	No Breach	GPD	2022-06-30	2022-09-30
	-12.7419%	-11.7982%	-13.9618%	-19.6750%	3.3097%	1.1461%	-1.9774%	GPD	2022-09-30	2022-12-31
	-6.0058%	-11.7219%	-14.0687%	-19.5278%	No Breach	No Breach	No Breach	Normal, t, GPD	2022-12-31	2023-03-31
	-6.1672%	-11.5208%	-13.9265%	-20.0038%	No Breach	No Breach	No Breach	Normal, t, GPD	2023-03-31	2023-06-30
	-4.5742%	-11.3230%	-13.7115%	-19.9972%	No Breach	No Breach	No Breach	Normal, t, GPD	2023-06-30	2023-09-30
	-4.8601%	-11.1144%	-13.7570%	-19.7355%	No Breach	No Breach	No Breach	Normal, t, GPD	2023-09-30	2023-12-31
	-6.3065%	-10.9257%	-13.5472%	-18.9109%	No Breach	No Breach	No Breach	Normal, t, GPD	2023-12-31	2024-03-31
-7.1625%	-10.7841%	-13.2632%	-19.0619%	-1.9651%	-4.4443%	No Breach	GPD	2024-03-31	2024-06-30	

-10.8060%	-10.6753%	-13.1550%	-18.7930%	4.3600%	1.8803%	2.5348%	t	2024-06-30	2024-09-30
-7.2426%	-10.6345%	-13.0846%	-18.9845%	-2.1205%	-4.9740%	No Breach	GPD	2024-09-30	2024-12-31
-11.4002%	-10.5492%	-12.9699%	-20.1501%	2.3013%	-0.1194%	-5.2312%	GPD	2024-12-31	2025-03-31
-11.8028%	-10.5412%	-12.7784%	-18.3919%	1.2616%	-0.9756%	-3.7261%	GPD	2025-03-31	2025-06-30

F BTC ES Results - Quarterly

Empirical ES (Test)	Normal ES	T- Distribution ES	GPD ES	Mean Excess (Normal)	Mean Excess (t)	AMean Excess (GPD)	Best ES Method	Start	End
Continued on next page(s)									
-6.7042%	-17.0273%	-103.2683%	-32.4508%	No Breach	No Breach	No Breach	Normal, t, GPD	2012-09-30	2012-12-31
-4.8557%	-16.2340%	-359.8861%	-37.6710%	No Breach	No Breach	No Breach	Normal, t, GPD	2012-12-31	2013-03-31
-31.5719%	-15.5045%	-106.2215%	-28.3010%	11.7015%	-79.0155%	-1.0950%	t	2013-03-31	2013-06-30
-10.9440%	-16.6126%	-89.5530%	-47.8244%	-0.0990%	-73.0393%	-31.3107%	t	2013-06-30	2013-09-30
-34.0882%	-16.1635%	-81.7420%	-42.2238%	11.5354%	-56.2526%	-11.6304%	t	2013-09-30	2013-12-31
-15.3446%	-17.3256%	-76.9332%	-42.6027%	0.5604%	-60.5541%	-20.4949%	t	2013-12-31	2014-03-31
-9.8511%	-17.2212%	-72.5271%	-50.3464%	No Breach	-59.6027%	No Breach	Normal, GPD	2014-03-31	2014-06-30
-7.0861%	-16.8547%	-65.8473%	-43.9234%	No Breach	No Breach	No Breach	Normal, t, GPD	2014-06-30	2014-09-30
-6.0522%	-16.5031%	-63.5619%	-50.0521%	No Breach	No Breach	No Breach	Normal, t, GPD	2014-09-30	2014-12-31
-16.1166%	-16.1746%	-53.5661%	-36.8505%	1.8524%	-35.5390%	-11.4330%	t	2014-12-31	2015-03-31
-4.3191%	-16.0132%	-52.0711%	-35.4612%	No Breach	No Breach	No Breach	Normal, t, GPD	2015-03-31	2015-06-30
-6.9186%	-15.6468%	-53.2873%	-31.6031%	No Breach	No Breach	No Breach	Normal, t, GPD	2015-06-30	2015-09-30
-9.3011%	-15.3343%	-55.4563%	-33.2931%	No Breach	-43.4864%	No Breach	Normal, GPD	2015-09-30	2015-12-31
-6.5245%	-15.0426%	-50.2321%	-34.0744%	No Breach	-39.2819%	No Breach	Normal, GPD	2015-12-31	2016-03-31
-6.2818%	-14.7692%	-52.6845%	-31.9385%	-1.9587%	-39.8740%	No Breach	GPD	2016-03-31	2016-06-30
-6.4984%	-14.5247%	-58.4397%	-33.9447%	No Breach	No Breach	No Breach	Normal, t, GPD	2016-06-30	2016-09-30
-2.5926%	-14.2739%	-69.0702%	-33.8721%	No Breach	No Breach	No Breach	Normal, t, GPD	2016-09-30	2016-12-31

-10.7880%	-13.9956%	-72.9206%	-38.0115%	-1.0361%	-59.9611%	-23.7785%	t	2016-12-31	2017-03-31
-9.7628%	-13.8576%	-63.2252%	-28.4054%	0.4227%	-48.9449%	No Breach	GPD	2017-03-31	2017-06-30
-11.1170%	-13.6872%	-51.9780%	-28.3416%	-1.5794%	-39.8702%	No Breach	GPD	2017-06-30	2017-09-30
-8.4488%	-13.6213%	-46.4348%	-27.9229%	No Breach	-36.8306%	No Breach	Normal, GPD	2017-09-30	2017-12-31
-17.3272%	-13.5204%	-41.0449%	-27.6958%	2.4996%	-27.4101%	-5.2439%	t	2017-12-31	2018-03-31
-9.5088%	-13.5820%	-38.1091%	-26.9752%	-1.2755%	-26.8749%	No Breach	GPD	2018-03-31	2018-06-30
-7.1086%	-13.4586%	-35.1312%	-26.9891%	No Breach	No Breach	No Breach	Normal, t, GPD	2018-06-30	2018-09-30
-12.5745%	-13.3197%	-33.2232%	-27.0807%	0.1710%	-20.6487%	-13.0197%	t	2018-09-30	2018-12-31
-5.7788%	-13.2599%	-33.5911%	-26.9136%	No Breach	-23.8089%	No Breach	Normal, GPD	2018-12-31	2019-03-31
-11.3207%	-13.0987%	-34.9330%	-26.7056%	1.2546%	-22.4115%	-9.1751%	t	2019-03-31	2019-06-30
-12.4482%	-13.0354%	-34.1125%	-26.4229%	0.3763%	-21.6644%	-12.3220%	t	2019-06-30	2019-09-30
-6.1802%	-12.9838%	-32.9356%	-26.6109%	No Breach	No Breach	No Breach	Normal, t, GPD	2019-09-30	2019-12-31
-16.0156%	-12.8695%	-32.7121%	-26.2811%	6.1427%	-13.6999%	-2.8900%	t	2019-12-31	2020-03-31
-7.8172%	-12.8646%	-31.4851%	-26.5614%	1.7865%	-16.8340%	-11.9102%	t	2020-03-31	2020-06-30
-5.2040%	-12.7514%	-30.6176%	-26.4459%	No Breach	No Breach	No Breach	Normal, t, GPD	2020-06-30	2020-09-30
-4.6256%	-12.6221%	-29.6868%	-26.8941%	-2.4554%	-19.5201%	No Breach	GPD	2020-09-30	2020-12-31
-12.0138%	-12.4884%	-28.9524%	-26.2434%	1.2539%	-15.2101%	-11.2248%	t	2020-12-31	2021-03-31
-11.1945%	-12.4394%	-27.7549%	-25.5422%	-0.7838%	-17.5237%	-13.0439%	t	2021-03-31	2021-06-30
-7.3302%	-12.4224%	-26.4047%	-26.0846%	-2.0526%	-16.0349%	No Breach	GPD	2021-06-30	2021-09-30
-7.0562%	-12.3369%	-24.9932%	-26.1458%	No Breach	-16.4203%	No Breach	Normal, GPD	2021-09-30	2021-12-31
-8.5274%	-12.2590%	-24.0645%	-26.0210%	-0.3648%	-12.1703%	-14.1269%	GPD	2021-12-31	2022-03-31
-13.9278%	-12.1903%	-23.5272%	-25.7327%	6.9006%	-9.5993%	-6.6418%	t	2022-03-31	2022-06-30
-8.5293%	-12.1843%	-23.1948%	-25.1981%	-2.0035%	-13.3069%	No Breach	GPD	2022-06-30	2022-09-30

-9.9206%	-12.1063%	-22.7971%	-25.1929%	1.5788%	-9.1120%	-7.9131%	t	2022-09-30	2022-12-31
-5.4493%	-12.0379%	-22.8045%	-25.0028%	No Breach	-14.3680%	No Breach	Normal, GPD	2022-12-31	2023-03-31
-5.6276%	-11.9539%	-22.5760%	-25.3558%	No Breach	No Breach	No Breach	Normal, t, GPD	2023-03-31	2023-06-30
-4.7321%	-11.8656%	-21.9340%	-25.3229%	No Breach	No Breach	No Breach	Normal, t, GPD	2023-06-30	2023-09-30
-4.4400%	-11.7735%	-22.0313%	-25.6849%	No Breach	-14.1595%	No Breach	Normal, GPD	2023-09-30	2023-12-31
-5.7373%	-11.6785%	-21.5085%	-24.9605%	No Breach	No Breach	No Breach	Normal, t, GPD	2023-12-31	2024-03-31
-6.1780%	-11.6022%	-22.0854%	-25.1426%	No Breach	No Breach	No Breach	Normal, t, GPD	2024-03-31	2024-06-30
-7.0731%	-11.5348%	-20.2833%	-25.5467%	2.4366%	-6.3119%	-11.5753%	GPD	2024-06-30	2024-09-30
-5.0121%	-11.4745%	-19.8689%	-24.9535%	No Breach	No Breach	No Breach	Normal, t, GPD	2024-09-30	2024-12-31
-6.2277%	-11.4015%	-19.4382%	-25.3499%	No Breach	-10.9646%	No Breach	Normal, GPD	2024-12-31	2025-03-31
-5.2151%	-11.3372%	-19.0184%	-24.3480%	No Breach	No Breach	No Breach	Normal, t, GPD	2025-03-31	2025-06-30